



Scientific Committee on Consumer Safety

SCCS

SCIENTIFIC ADVICE ON

Butylated Hydroxyanisole (BHA)



The SCCS adopted this document during the plenary meeting on 30 October

ACKNOWLEDGMENTS

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1. ABSTRACT

The SCCS concludes the following:

In light of technical and scientific progress, available scientific literature, relevant in silico tools and the SCCS' expert judgement and taking under consideration in particular the concerns related to potential endocrine disrupting properties, the SCCS is requested:

1. *to identify and justify whether there are specific concerns regarding the use of BHA in cosmetic products;*

The SCCS considers the use of BHA in leave-on and rinse off cosmetic products up to a concentration of 0.07% as safe.

2. *to highlight if there is a potential risk for human health from the use of BHA in cosmetic products;*

This SCCS Opinion considered only dermal use and hence, is applicable for the use of BHA in dermally applied products, and not in oral care products or cosmetic products which may lead to exposure of the end-user's lungs by inhalation.

The SCCS mandate does not address environmental aspects. Therefore, this assessment has not covered the safety of BHA for the environment.

Keywords: SCCS, scientific advice, Butylated Hydroxyanisole, BHA, Regulation 1223/2009, CAS No. 25013-16-5, EC No. 246-563-8.

Opinion to be cited as: SCCS (Scientific Committee on Consumer Safety), scientific advice on the safety of Butylated Hydroxyanisole (BHA) (CAS No. 25013-16-5, EC No. 246-563-8) as a substance with potential endocrine disrupting properties in cosmetic products, preliminary version of 30 October 2025, SCCS/1682/25

Scientific advice on the safety of Butylated Hydroxyanisole (BHA) as a substance with potential endocrine disrupting properties in cosmetic products

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These Committees are: the Scientific Committee on Consumer Safety (SCCS) and the Scientific Committee on Health, Environmental and Emerging Risks (SCHEER) and they are made up of scientists appointed in their personal capacity.

In addition, the Commission relies upon the work of the European Food Safety Authority (EFSA), the European Medicines Agency (EMA), the European Centre for Disease prevention and Control (ECDC) and the European Chemicals Agency (ECHA).

SCCS

The Committee shall provide Opinions on questions concerning health and safety risks (notably chemical, biological, mechanical and other physical risks) of non-food consumer products (for example cosmetic products and their ingredients, toys, textiles, clothing, personal care and household products such as detergents, etc.) and services (for example: tattooing, artificial sun tanning, etc.).

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2. MANDATE FROM THE EUROPEAN COMMISSION

Substances with endocrine disrupting properties

On 7 November 2018, the Commission adopted a review¹ of Regulation (EC) No 1223/2009 on cosmetic products ('Cosmetics Regulation') regarding substances with endocrine disrupting properties. The review concluded that the Cosmetics Regulation provides the adequate tools to regulate the use of cosmetic substances that present a potential risk for human health, including when displaying ED properties.

The Cosmetics Regulation does not have explicit provisions on EDs. However, it provides a regulatory framework with a view to ensuring a high level of protection of human health. Environmental concerns that substances used in cosmetic products may raise are considered through the application of Regulation (EC) No 1907/2006 ('REACH Regulation').

In the review, the Commission commits to establishing a priority list of potential EDs not already covered by bans or restrictions in the Cosmetics Regulation for their subsequent safety assessment. A priority list of 28 potential EDs in cosmetics was consolidated in early 2019 based on input provided through a stakeholder consultation. The Commission then organised a public call for data² from 16 May 2019 to 15 October 2019 on 14³ of the 28 substances (to be treated with higher priority) in order to be able to prepare the safety assessment of these substances. Triclocarban and Triclosan are among the above-mentioned 14 substances for which the call for data took place.

Existing information on Butylated Hydroxyanisole

The ingredient Butylated Hydroxyanisole (INCI: BHA) (CAS No. 25013-16-5, EC No. 246-563-8) with the chemical name '2-tert-butyl-4-methoxyphenol' is a monohydric phenolic antioxidant. BHA is a mixture of two isomers, 3-tertiary-butyl-4-hydroxyanisole and 2-tertiary-butyl-4-hydroxyanisole.

BHA is currently not regulated under the Cosmetics Regulation (EC) No. 1223/2009; however, it has been used for decades in various industries, including personal care, pharma, and food industry as an antioxidant to extend the shelf life of products.

According to CosIng database, BHA is reported to have the following functions: 'antioxidant' and 'fragrance'. Cosmetic products often contain synthetic antioxidants like BHA to prevent the oxidation of fats and oils (mostly essential oils). In particular, the conjugated aromatic ring of BHA can stabilize free radicals, sequestering them acting as free radical scavenger.

BHA is closely related to BHT (butylated hydroxytoluene), which has been recently evaluated by the SCCS in view of its potential endocrine activity. In that Opinion⁴, the SCCS concluded on the safety (under certain conditions) of BHT, when used in cosmetic products. BHA is also suspected of being an endocrine disruptor.

Given the possible health risks associated with this ingredient, its safe use in cosmetic products should be evaluated by the SCCS in view of technical and scientific progress. The Commission, therefore, requests the SCCS to provide a scientific advice on the safety of BHA when used in cosmetic products.

¹ <https://ec.europa.eu/transparency/regdoc/rep/1/2018/EN/COM-2018-739-F1-EN-MAIN-PART-1.PDF>

² https://ec.europa.eu/growth/content/call-data-ingredients-potential-endocrine-disrupting-properties-used-cosmetic-products_en

³ Benzophenone-3, kojic acid, 4-methylbenzylidene camphor, propylparaben, triclosan, resorcinol, octocrylene, triclocarban, butylated hydroxytoluene (BHT), benzophenone, homosalate, benzyl salicylate, genistein and daidzein

⁴ SCCS (Scientific Committee on Consumer Safety), scientific opinion on Butylated hydroxytoluene (BHT), preliminary version of 27 September 2021, final version of 2 December 2021, SCCS/1636/21

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disrupting properties in cosmetic products

Terms of reference

In light of technical and scientific progress, available scientific literature, relevant in silico tools and the SCCS' expert judgement and taking under consideration in particular the concerns related to potential endocrine disrupting properties, the SCCS is requested:

1. *to identify and justify whether there are specific concerns regarding the use of BHA in cosmetic products;*
2. *to highlight if there is a potential risk for human health from the use of BHA in cosmetic products;*

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3. SCIENTIFIC ADVICE

3.1 CHEMICAL AND PHYSICAL SPECIFICATIONS

3.1.1 Chemical identity

3.1.1.1 Primary name and/or INCI name

2-*tert*-Butyl-4-methoxyphenol

3.1.1.2. Chemical names

IUPAC Name:

BHA: 2-*tert*-butyl-4-methoxyphenol.

BHA is composed of the following two isomers:

- 2-BHA: 3-(*tert*-Butyl)-4-methoxyphenol
(other synonyms: Phenol, 3-(1,1-dimethylethyl)-4-methoxy-, Phenol, 3-*tert*-butyl-4-methoxy-, p-methoxy-m-*tert*-butylphenol, 3-(1,1-Dimethylethyl)-4-methoxyphenol, 2-(1,1-Dimethylethyl)-4-hydroxyanisole, and several others)
- 3-BHA: 3-*tert*-Butyl-4-hydroxyanisole
(other synonyms: 4-Hydroxy-3-*tert*-butylanisole, 3-t-Butyl-4-hydroxyanisole, 3-*tert*-Butyl-p-hydroxyanisole, Phenol, 2-(1,1-dimethylethyl)-4-methoxy-, 4-Methoxy-2-*tert*-butylphenol, and several others)

2-*tert*-butyl-4-methoxyphenol

Computed by Lexichem TK 2.7.0 (PubChem release 2021.10.14)

From <https://pubchem.ncbi.nlm.nih.gov/compound/8456#section=Names-and-Identifiers>,
consulted on 27th September 2024

3-*tert*-Butyl-4-hydroxyanisole

4-Hydroxy-3-*tert*-butylanisole

2-(*tert*-butyl)-4-methoxyphenol

3-BHA

3-T-BUTYL-4-HYDROXYANISOLE

3-*tert*-Butyl-p-hydroxyanisole

Phenol, 2-(1,1-dimethylethyl)-4-methoxy-

4-Methoxy-2-*tert*-butylphenol

And several others

From <https://pubchem.ncbi.nlm.nih.gov/compound/8456#section=Depositor-Supplied-Synonyms>,
consulted on 27th September 2024

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3.1.1.3 Trade names and abbreviations

3.1.1.4 CAS / EC number

CAS No. 25013-16-5,
EC No. 246-563-8

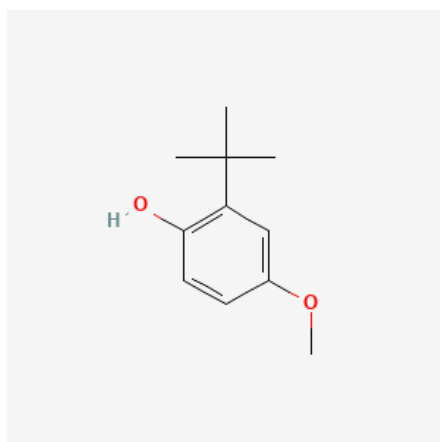
SCCS comment

According to Anses (2024), in 2015 the substance was registered under REACH as *tert*-butyl-4-methoxyphenol with the identifiers EC 246-563-8, CAS 25013-16-5 and composed of two isomers i.e. 2-*tert*-butyl-4-methoxyphenol (CAS 121-00-6, EC 204-442-7) and 3-*tert*-butyl-4-methoxyphenol (CAS 88-32-4, EC 201-820-3). The ratio of the two isomers was however not specified. In 2021, the identity of the substance was established to be the isomer 2-*tert*-butyl-4-methoxyphenol (3-BHA), CAS 121-00-6, EC 204-442-7 (i.e. EC 246-563-8 comprises 80% or more EC 204-442-7).

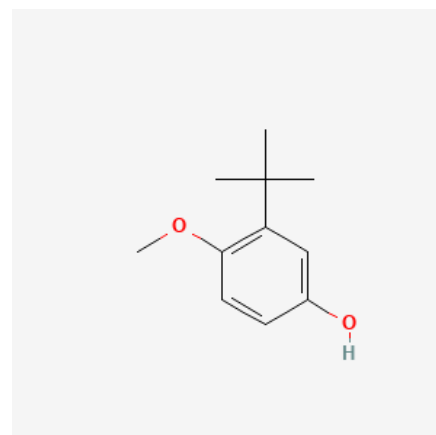
For this Scientific Advice, information retrieved from literature searches for all three CAS numbers mentioned above (i.e. CAS 88-32-4, CAS 121-00-6 and CAS 25013-16-5) was used. Throughout this Opinion, the SCCS will use the abbreviation BHA for either of the CAS or EC numbers given above and further specify, if considered necessary.

Ref.: Anses, 2024

3.1.1.5 Structural formula



3-BHA



2-BHA

From <https://pubchem.ncbi.nlm.nih.gov/compound/8456#section=Structures>, consulted on 27th September 2024

3.1.1.6 Empirical formula

C₁₁H₁₆O₂

3.1.2 Physical form

Butylated hydroxyanisole appears as white, beige or slightly yellow waxy solid with an aromatic odor and a slightly bitter burning taste. (NTP, 1992).

From National Toxicology Program, Institute of Environmental Health Sciences, National Institutes of Health (NTP). 1992. National Toxicology Program Chemical Repository Database. Research Triangle Park, North Carolina, cited by

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<https://pubchem.ncbi.nlm.nih.gov/compound/8456#section=Experimental-Properties>,
consulted on 27th September 2024

3.1.3 Molecular weight

180.24 g/mol, Computed by PubChem 2.2 (PubChem release 2021.10.14).

From <https://pubchem.ncbi.nlm.nih.gov/compound/8456#section=Chemical-and-Physical-Properties>, consulted on 27th September 2024

3.1.4 Purity, composition and substance codes

Regarding the use of BHA as food and feed additives and food contact material, the additive BHA is considered a mixture of 2-*tert*-butyl-4-hydroxyanisole and 3-*tert*-butyl-4-hydroxyanisole. It is specified to contain at least 98.5% of BHA (IUPAC name: 2-*tert*-butyl-4-hydroxyanisole and 3-*tert*-butyl-4-hydroxyanisole, Chemical Abstracts Service (CAS) number 25013-16-5, chemical formula C₁₁H₁₆O₂, and molecular weight 180.25) and not less than 85% of the 3-*tert*-butyl-4-hydroxyanisole isomer. In a more recent publication, it is indicated that the commercial product is a mixture, typically consisting of 10% 2-BHA and 90% 3-BHA.

Ref.: EFSA, 2011; Sun et al., 2019

3.1.5 Impurities / accompanying contaminants

The following impurities have been reported for BHA as it is used in cosmetics: 4-hydroxyanisole (0.5% maximum), 1-*t*-butyl-2,5-dimethoxybenzene (0.5% maximum), 2,5-di-*t*-butyl-hydroxyanisole (0.2% maximum), hydroquinone dimethyl ether (0.1% maximum), sulfated ash (0.01% maximum), lead (as Pb; 20 ppm maximum), and arsenic (as As; 3 ppm maximum).

Ref.: Elder, 1984.

3.1.6 Solubility

Water solubility

610 mg/l at 29 °C and pH 1.28 (shake-flask method, OECD TG 105).

Ref. Anses, 2024, ECHA CHEM database

The solubility in acetonitrile and methanol was found to be greater than 1000 mg/L, while in acetone, ethanol, and toluene, it ranged from 200 to 1000 mg/L. In DMF, the solubility was between 100 and 200 mg/L, all measured at 29.4°C.

Ref.: ECHACHEM database

Solubilities in other media

Soluble in petroleum ether, ethanol

Ref.: Haynes 2014-2015

Freely soluble in alcohol, propylene glycol, chloroform, ether; soluble in petroleum ether (Shellysolve H), fats, oils.

Ref.: O'Neil, 2013

1% in glycerine at 100 °C

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Ref.: Lefaux, 1968

From References cited by
<https://pubchem.ncbi.nlm.nih.gov/compound/8456#section=Solubility>,
consulted on 2nd September 2024

3.1.7 Partition coefficient (Log Pow)

Log Pow: 2.82 at 25°C and pH 7.5 (HPLC method, OECD TG 117)

Ref.: Anses, 2024, ECHA CHEM database

3.1.8 Additional physical and chemical specifications

Organoleptic properties (colour, odour, taste if relevant):

White solid at 20 °C and 101.3 kPa

Ref.: Anses, 2024

Melting point

57-59°C at 977.0 hPa (OECD TG 102) (Anses, 2024, ECHA, 2013)

Boiling point:

- > 240°C at 980.2 hPa (OECD TG 103) (Anses, 2024, ECHA, 2013)

Flash point:

- 116.6 °C at 981.0 hPa (EU Method A.9)

Ref.: Anses, 2024, ECHA, 2013

Vapour pressure:

0.311 Pa at 25°C (Modified Grain Method, QSAR-EPI Suite). Indicative value, no reliable data available (Anses, 2024, ECHA CHEM database)

Density:

Bulk density: 0.6896 g.cm⁻³ at 28.6 °C

Tap density: 0.6896 g/cm⁻³ at 28.6 °C

Ref.: Anses, 2024, ECHA, 2013

pKa=0.000167333 at 27°C (conductometric method similar to OECD TG 112)

Ref.: Anses, 2024, ECHA, 2013

pH (1 vol%): 4.78 at 25 °C (OECD 122)

Ref.: ECHA, 2013

3.1.9 Homogeneity and Stability

According to Safety Data Sheets of various companies, BHA is stable under normal ambient and anticipated storage and handling conditions of temperature and pressure.

BHA was observed to be stable in dichloromethane, with no degradation products detected after 24 hours, as confirmed by the GC-MS chromatograms taken at 0 and 24 hours.

Ref.: ECHACHEM database

BHA undergoes degradation when exposed to sunlight or UV. Degradation products following a 24-hour exposure of BHA to UV radiation included 2-tert-butyl-quinone; 2-tert-butyl-hydroquinone; 2-tert-butyl-1,4-dimethoxybenzene; 2',3-di-tert-butyl-2-hydroxy-4',5-dimethoxydiphenyl ether; 3,3'-di-tert-butyl-2,2'-dihydroxy-5,5'-dimethoxy biphenyl; 3,3'-di-tert-butyl-2'-hydroxy-2,5,5'-trimethoxy biphenyl; and 3,3'-di-tert-butyl-2,2', 5,5'-tetramethoxybiphenyl.

Ref.: Kurechi, 1977

BHA was irradiated in soybean oil for 70 days and in lard for 40 days under sunlight, affording 3, 3'-di-tert-butyl-2,2'-di-hydroxy-5,5'-di-methoxy Biphenyl (A) and 2', 3-di-tert-butyl-2-hydroxy-4',5-di-methoxy Biphenyl ether (B). Under high concentration (5%) of BHA, compound A was more formed than B, and under low concentration (0.5%), B was more formed than A. Rapid rancidity of fat and oil occurred after disappearance of BHA and photolytic products (A, B).

Ref.: Mihara, 1974

3.2 TOXICOKINETICS

The tissue-distribution and kinetics of orally administered labelled [methyl-¹⁴C]3-BHA were investigated in 39 adult male Sprague-Dawley rats. [Methyl-¹⁴C]3-BHA was administered by gavage as a single dose (1.5 mmol/kg bw, equivalent to 270 mg/kg bw). Urine, faeces, blood and samples from all major organs were collected at 0.5, 1, 3, 6, 12, 16, 17, 18, 24, 48, 72, 168 and 240 hours after dosing. The total radiolabel in most tissues increased exponentially with time, reaching a maximum between 10-24 hours after dosing, followed by an exponential decrease. By 48 hours, the radioactivity was almost totally eliminated from the rats with 41% of the methoxy-labelled radioactivity found in the urine and 53% in faeces.

Ref.: Ansari and Hendrix, 1985, quoted from EFSA (2011).

The elimination of BHA (isomers of 2-BHA and 3-BHA labelled at either the methoxy or tert-butyl moiety) was studied in 2 groups of 3 male F344 rats. BHA given by gavage as a single dose of 1 g/kg bw was found to be excreted via urine, faeces or respiration at a level of 87-96% within the first 48 hours after administration. Animals were dosed with one of the four compounds: 2- and 3-BHA labelled at either the methoxy or tert-butyl moiety. The two isomers of BHA, 3-BHA and 2-BHA, with the tert-butyl label, were found in the urine at levels of 63.7% and 69.0% of the dose of the corresponding isomers, respectively. In faeces, 28.8% and 18.1% of the dose of the corresponding isomers were found for 3-BHA and 2-BHA, with the tert-butyl label, respectively. 3-BHA and 2-BHA, labelled in the methoxy group, were found in the urine at levels of 49.8% and 46.5% of the dose of the corresponding isomer, respectively, while 28.3% and 29.6% of the dose of the corresponding isomer were found in faeces for 3-BHA and 2-BHA, labelled in the methoxy group. In expired air, 8.3% and 13.7% of the dose of 2-BHA and 3-BHA, labelled in the methoxy group, were found.

Ref.: Hirose *et al.* (1987a), quoted from EFSA (2011)

The elimination pattern of BHA was investigated in four groups of three male F344 rats (6 weeks old, weighing 120-130 g), each given by gavage a single dose (1 g/kg bw) of one of the following labelled preparations: [tert-butyl-¹⁴C]3-BHA, [butyl-¹⁴C]2-BHA, [methyl-¹⁴C]3-BHA or [methyl-¹⁴C]-2-BHA). The preparations were obtained by diluting the radio-labelled molecules with unlabelled 3-BHA and 2-BHA. By using thin-layer chromatography (TLC), the urinary and faecal metabolites were identified in samples of 2-day pooled urine and faeces, and further characterized and identified by enzymatic hydrolysis. Confirmation of the standards was performed by proton magnetic resonance spectroscopy and electron impact mass spectrometry. The radioactivity was almost completely eliminated [within two days] and the total ¹⁴C recoveries were about 92% in 3 male rats. The total ¹⁴C recoveries from [methyl-¹⁴C]3-BHA and [methyl-¹⁴C]-2-BHA were lower than those from [tert-butyl-¹⁴C]3-BHA and [butyl-¹⁴C]2-BHA, being 81.3 and 73%, respectively. Most of the metabolites of 3-BHA and 2-BHA were found as conjugates (53% of 3-BHA and 66% of 2-BHA). The major metabolite identified in the urine of rats given 3-BHA was 3-BHA-glucuronide. A smaller amount of tert-butylhydroquinone-sulphate was detected in the faeces together with unchanged 3-BHA and 3-BHA-glucuronide. The major metabolites identified in the urine of rats given 2-BHA were the sulphate conjugates of 2-BHA, 4-tertbutyl-5-methoxy-1, 2-benzoquinone and the glucuronide of 2-BHA. Unchanged 2-BHA was found in the faeces.

Ref.: Hirose *et al.* (1988), quoted from EFSA (2011)

The absorption and excretion of [methyl-¹⁴C]3-BHA was investigated in 9 male beagle dogs (5 months of age). Groups of 3 dogs were fed a diet containing 0, 0.03 or 3% BHA for 7 days followed by a single injection of radiolabelled [3-]BHA. Radioactivity was determined in whole body, blood, urine, faeces and several tissues including different parts of the stomach. Within 48 hours 50-80% of the labelled [3-]BHA was detected in the urine and 15-30% in faeces. Seven days after administration the levels of radiolabelled [3-]BHA were measured in organs and showed a small amount in the stomach (0.16-0.19% of dose/g), liver (0.3-1.7%) and other tissues (0.02%). It was suggested by the authors, that the dose-independence of distribution and excretion of radiolabelled [3-]BHA could be attributed to the lack of a forestomach in the beagle dog.

Ref.: Takizawa *et al.* (1985), quoted from EFSA (2011)

The metabolism of 2- and 3-BHA was investigated in rats, focusing on the forestomach and binding to tissue macromolecules (Hirose *et al.*, 1987b). Male Fischer 344 rats (2 groups of 3 rats) were given by gavage a single dose of 1 g/kg bw [tert-butyl-¹⁴C]3-BHA or [methyl-¹⁴C]3-BHA. BHA metabolites were analysed by comparing TLC of forestomach, glandular stomach and stomach content samples, taken 6 hours after administration. No significant amount of metabolites was found in the forestomach or glandular stomach epithelium. In the binding experiment, rats were given by gavage, a single dose of 1 g/kg bw of [tert-butyl-¹⁴C]3-BHA, [butyl-¹⁴C]2-BHA or [methyl-¹⁴C]3-BHA with or without pretreatment with 1% unlabelled 3-BHA in the diet for 6 days. Based on the data, it was suggested by the authors that BHA acts on forestomach epithelium without metabolic activation, and furthermore that the action was not related to its binding to DNA or RNA.

Ref.: Hirose *et al.* (1987b), quoted from EFSA (2011)

Transplacental transfer

In mice, Ahmed *et al.* (1991), investigated the placental transport and localization in fetal and maternal tissues of [methyl-¹⁴C]2-BHA and [methyl-¹⁴C]3-BHA, and showed differences between 3-BHA and 2-BHA. Pregnant mice were given BHA isomers i.v. at a dose level of 50 µCi/100 g at day 11 and day 18 of gestation and distribution was investigated by whole-body autoradiography. Peak levels of radioactivity occurred in various tissues at 1-4 hours after administration of both isomers. The tissue distribution and affinity was shown to be different for the two isomers including the magnitude and extent of placental transport. 3-BHA levels in maternal liver and brown fat tissue were much higher than the radioactivity concentration in the corresponding tissues of mothers treated with 2-BHA. Fetal concentration was higher in animals treated with 2-BHA than in those treated with 3-BHA. Thus, a different placental transport pattern was shown for radiolabelled 2-BHA and 3-BHA, with 2-BHA being more accumulated in fetal tissue and 3-BHA being more accumulated in maternal tissue. Both isomers were shown to cross the placental membrane.

Ref.: EFSA, 2011

Groups of seven pregnant Wistar rats were given 0, 200, 400 or 800 mg of 3-tert-butyl-4-hydroxyanisole (BHA) dissolved in sesame oil by gavage from GD 6 to GD 15. The concentration of conjugated BHA in the fetuses was significantly lower than in the liver and serum of the mothers, while the concentration of free BHA in the serum of the mothers was as high as in the fetuses.

Ref.: Yamada *et al.* (1993)

Metabolism *in vivo*

Based on *in vivo* studies performed in rats, rabbits, dogs and humans, the main metabolic pathways are either glucuronidation or sulfation of the free hydroxyl group with species differences in the amount of these conjugation reactions: In rats, rabbits and in man, glucuronidation predominates while in dogs, sulfation is the preferred route. Furthermore, in

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rats, 3-BHA was predominantly glucuronidated while 2-BHA was mainly sulphated. O-demethylation was reported as a minor pathway in rats, rabbits and dogs.

Ref.: Conning and Philips (1986), quoted from ANSES (2024)

In humans, the elimination and distribution of BHA (as a mixture of isomers containing 93% 3-BHA) was investigated in eight healthy male non-smoking volunteers who ingested 0.5 mg BHA/kg bw/day for 10 consecutive days (Verhagen *et al.*, 1989). Standard biochemical plasma parameters, kinetic parameters and urinary excretion of metabolites (total, conjugated and unconjugated BHA) were determined. No significant differences in kinetic parameters or urinary metabolite profiles were detected. Significantly higher amounts of unconjugated BHA and TBHQ were detected in the urine on days 3 and 7 compared to the first day of BHA administration. The study authors concluded that this may result either from an induction or inhibition of BHA-specific phase-I and -II metabolizing enzymes in man, or from a bioaccumulation of BHA and/or its metabolites in the body.

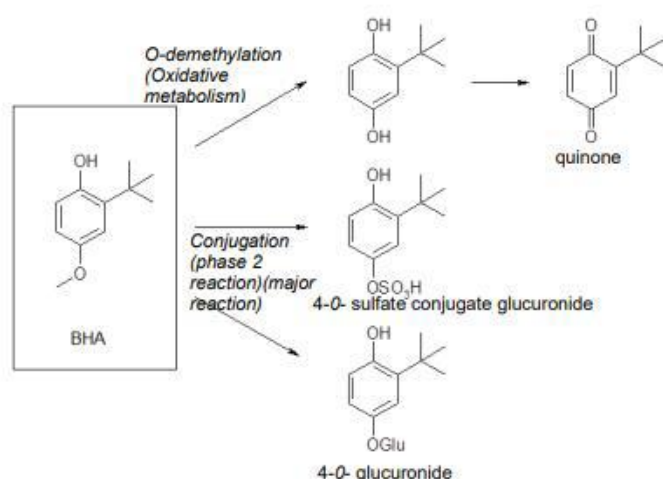
Ref.: Verhagen *et al.*, 1989, quoted from EFSA (2011)

Metabolism *in vitro*

The metabolism of 3-BHA was studied by incubating 3-BHA with liver microsomes from rats given β -naphthoflavone by p.o. intubation. Three metabolites were identified, two major metabolites and a minor metabolite. One of the major metabolites was the catechol of 3-BHA, i.e., 3-ter-butyl-4,5-dihydroxyanisole. A characteristic of this compound is its capacity to be oxidized readily. The second major metabolite was tert-butyl hydroquinone. The third metabolite, which occurred in small quantities, was 2,2'-dihydroxy-3,3'-di-tert-butyl-5,5'-dimethoxydiphenyl. The results obtained in this study were confirmed by Cummings *et al.* 1985.

Ref.: Armstrong and Wattenberg, 1985, quoted from ANSES (2024).

Figure 1: proposed metabolic pathways for BHA (taken from ANSES, 2024):



3.2.1 Dermal / percutaneous absorption

Dermal absorption of BHA was investigated *in vitro* in order to assess systemic uptake after cosmetic use. Human abdominal skin from 5 donors (2 males, 3 females) was obtained 24 hr post mortem, stored in a refrigerator and used within 12 hr. BHA and [^{14}C]-BHA were added to a water-in-oil ointment to prepare ointments containing 1% or 0.07% BHA. It was further mentioned that cosmetic products may contain up to 0.07% BHA. 10 mg of the ointment was applied on a skin surface area of 1.77 cm², inserted in a penetration chamber and left for either 4 (1% BHA) or 16 hours (0.07% BHA). Physiological saline was used as receptor fluid.

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After removal of the horny layer (10-20 tape strippings), skin sections, tapes and receptor fluids were analyzed for [^{14}C] content. After 4 hr and using a 1% BHA concentration, radioactivity was not detectable in the receptor fluid. However, after 16 hr and using the lower concentration, radioactivity was detectable in receptor fluid. The mean of 0.18 μg BHA (2.68%) penetrated into the receptor fluid, 30.5% remained on the skin surface, 6.7% was detected in the horny layers and 50.8% was detected in the remaining skin layers.

Table 1**Table I**

In Vitro Permeation of 0.07% BHA in a W/O Ointment (10 mg) Through Whole-Thickness Excised Human Skin (surface area 1.77 cm^2) After 16 h (6 ml physiological saline as acceptor medium)

	BHA equivalents in μg , (mean $\pm$ standard deviation, n = 3)	% Applied dose
♂, 75 yr	0.17 $\pm$ 0.044	2.2–3.0
♂, 90 yr	0.33 $\pm$ 0.120	3.2–7.1
♀, 58 yr	0.11 $\pm$ 0.055	0.7–2.0
♀, 81 yr	0.11 $\pm$ 0.012	1.5–1.8
♀, 24 yr	0.18 $\pm$ 0.140	1.6–5.3

BHA Equivalents in Three Parallel Samples of a Skin Specimen (♀, 24 yr)

	BHA equivalents (μg)	% Applied dose
Residue on the skin surface	2.02 1.75 1.96	31.60 29.30 30.70
Horny layer (10 strippings)	0.38 0.41 0.53	5.89 6.89 7.52
Whole-skin sample after stripping (solubilized in 10 N KOH, 4 h at 55–60°C)	3.28 3.23 3.27	51.30 54.20 46.90
Amount in the chamber fluid	0.10 0.06 0.37	1.56 1.02 5.34
Total recovery		90.40 91.40 90.50

Ref.: Schumann, 1991

SCCS comment

The study results indicate dermal penetration up to 5.3 % of a 0.07% concentration of BHA in a water/oil ointment. However, the results were from a publication described in public literature as lacking important details (e.g. composition of the ointment, details on position of the labelling). Furthermore, the study was not performed in line with the SCCS requirements (use of full thickness skin, exposure time of 16 hrs) and high variability in study results was observed. Therefore, the SCCS will use a default of 50% dermal absorption for the safety calculation.

3.2.2 Other studies on toxicokinetics

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SCCS overall comment on toxicokinetics

Overall, oral absorption of BHA is at least 86% in rats, 44% in dogs, and 32–100% in humans. The majority of the dose is excreted in the urine of both humans and rats during the first two days after oral administration (rat: approx. 49%, human: approx. 39%), primarily in conjugated form and as the demethylation product TBHQ or its conjugates (rat 8%, human 9%). Approximately 32% of free BHA is found in rat faeces; it is not detectable in humans. The plasma half-life of free BHA is less than one hour in both humans and rats. Excretion is largely complete after four days. BHA passes to rat fetuses to a small extent. Due to the high absorption rates observed in humans and animals, no correction for oral absorption will be made. For dermal absorption, a default of 50% is used for MoS calculation.

3.3 EXPOSURE ASSESSMENT

3.3.1 Function and uses

Cosmetic uses:

According to the CosIng database, BHA is reported to have the following functions: 'antioxidant' and 'fragrance'. Cosmetic products often contain synthetic antioxidants like BHA to prevent the oxidation of fats and oils (mostly essential oils). In particular, the conjugated aromatic ring of BHA can stabilise free radicals, sequestering them acting as free radical scavenger.

Other uses:

BHA is used in the formulation of pharmaceuticals, polymers, and personal care products, coating products and lubricants and greases. It is industrially used in plant protection products as coformulant.

According to CIR (2023; 2024), maximal concentrations of use in cosmetic products reported were 0.15% in 2023 with highest maximal concentrations reported for leave-on products (up to 0.15 %), whereas maximal concentrations reported for rinse-off products were up to 0.0084%. In general, maximal concentrations were lower in 2023 when compared to those reported in 2003. Further, the occurring frequency in cosmetic products was much lower in 2023 when compared to 2002 (70 versus 1224).

Ref.: CIR 2023; 2024

In a study investigating the presence of synthetic phenolic antioxidants in a total of 214 personal care products in Canada, BHA was identified in 4.2 % of the products (geometric mean concentration in personal care products: 3.93 ng/g, maximum: 1.3 µg/g).

Ref.: Liu and Marbury, 2019.

According to information obtained from German Cosmetic, Toiletry, Perfumery and Detergent Association (IKW e.V.), the intended use concentration in cosmetics is up 0.07% which is also supported in a publication specifically dedicated to assess dermal absorption from cosmetic use.

Ref. Schumann, 1991

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3.3.2 Calculation of SED/LED

Calculation of daily, relative daily and systemic exposure dose is based on *Table (5)* of the SCCS Notes of Guidance, 12th revision excluding oral care products.

<i>Type of cosmetic product exposure</i>	<i>Product category</i>	<i>Daily Exposure E_{product} (g/d)</i>	<i>Relative daily exposure $E_{\text{product}} / \text{bw}^1$ (mg/kg bw/d)</i>
<i>Rinse-off Skin & hair cleansing products</i>	Shower gel	0.19	2.79
	Hand wash soap	0.20	3.33
	Shampoo	0.11	1.51
	Hair conditioner	0.04	0.67
<i>Leave on Skin & hair cleansing products</i>	Body lotion	7.82	123.20
	Face cream	1.54	24.14
	Hand cream	2.16	32.70
	Deodorant non-spray	1.50	22.08
	Hair styling	0.40	5.74
<i>Make-up products</i>	Liquid foundation	0.51	7.90
	Make-up remover	0.50	8.33
	Lipstick	0.06	0.90
	Eye make-up	0.02	0.33
	Mascara	0.025	0.42
	Eyeliners	0.005	0.08
<i>Oral care Products²</i>	Toothpaste	0.14	2.16
	Mouthwash	2.16	32.54
Aggregate exposure		17.4	269

Considering a BHA content of 0.07% in the final product, this results in daily external BHA exposure of 0.164, 0.0058 and 0.158 mg/kg bw/d for rinse-off and leave-on products combined, rinse-off products and leave-on products, respectively.

Systemic exposure dose (SED):

A default value of 50% is taken for dermal absorption (see section 3.2.1). This results in SEDs of 0.082 mg/kg bw/d (for leave-on and rinse-off products combined), 0.0029 mg/kg bw/d for rinse-off products and 0.0079 mg/kg bw/d for leave-on products.

3.4 TOXICOLOGICAL EVALUATION

3.4.1. Irritation and corrosivity

3.4.1.1 Skin irritation

In an amended report issued by the Cosmetic Ingredient Review Panel (CIR) only studies performed with cosmetic products are listed. In an *in vivo* skin irritation study (OECD 404) performed in 2012 in White New Zealand Rabbits and described in the REACH registration dossier for EC 204-442-7, the substance was slightly irritating to rabbit skin. In an *in vitro* skin irritation study performed according to OECD TG 439 (reconstructed human epidermis test method), the substance was considered non-irritating to human skin (study reports not available for evaluation).

Ref.: CIR (2024); ECHA website

SCCS comment

BHA might be considered as slightly irritating to the skin.

3.4.1.2 Mucous membrane irritation / eye irritation

In the updated / amended CIR reports (2023; 2024), studies performed with cosmetic products were described. Based on studies performed with different cosmetic products, BHA was considered slightly irritating to the eyes. In an *in vitro* test performed according to OECD TG 492 (Reconstructed Human Cornea-like Epithelium (RhCE) Test), described in the REACH registration dossier for EC 204-442-7, the substance was considered non-irritating (study report not available for evaluation). An older study comparing three alternative tests with the Draize eye test, reported that BHA would be an extreme eye irritant.

Ref.: Blein *et al.*, 1991

SCCS comment

Reliable information on eye irritation potential of BHA is not available. However, concentration of BHA in cosmetic products is expected to be low and thus unlikely to result in eye irritation.

3.4.2 Skin sensitisation

According to the Applicant, reliable animal studies could not be located.

Information/case reports on skin sensitisation in humans have been compiled and reviewed in CIR (2023; 2024) and MAK (2012). CIR (2023) indicates that the incidence of contact dermatitis among 548 subjects patch tested with 2 percent BHA was reported by the North American Contact Dermatitis Group to be 2 percent (11 subjects).

From a compilation of available human information (case reports, clinical tests and studies with volunteers) (summarised below) MAK (2012) concluded that the skin sensitising potential of BHA is low.

BHA has been described as a probable cause of allergic contact dermatitis in the case of a positive patch test reaction to 2% BHA: sensitization to BHA in a cook with facial dermatitis (Gola *et al.* 1989); BHA as a possible ingredient in rubber gloves in a baker with pruritic, erythematous-vesicular hand dermatitis but simultaneous reaction to BHT and 2-tert-butyl-

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5-methylphenol (Rich *et al.* 1991); 2 patients with contact dermatitis caused by topical preparations (Tosti *et al.* 1987); BHA as a possible food additive in a patient with chronic lip swelling (Lewis *et al.* 1995); BHA in lipstick and an acyclovir cream (Baes and van Hecke 1990). Two patients with intolerance reactions to a topical hydrocortisone preparation containing BHA reacted to the pharmaceutical-grade BHA used in the cream (2% in Vaseline), but not to 2% BHA in a commercially available patch test preparation (Orton and Shaw 2001). Two reviews list several other older case reports of suspected sensitization to BHA as a component of cosmetic creams, lipsticks, topical preparations or as a component of mayonnaise (CIR 1984; Hausen *et al.* 1994).

In a North American study, a positive reaction to 2% BHA occurred in a total of three of 1191 patients tested between 1978 and 1980 (Mitchell *et al.* 1982). Between 1977 and 1983, a total of 713 of the 13,216 patients suspected of cosmetic intolerance were tested in twelve dermatological clinics. Three of those tested showed a positive reaction to BHA (n.d.; Adams and Maibach 1985). Between 1987 and 1989, 14 of 4,948 patients from 21 dermatological practices reacted positively to 2% BHA (Scheuer *et al.* 1992).

Of 376 patients tested between the ages of six months and twelve years, four showed a positive reaction to BHA (n.d.) (Conti *et al.* 1997). Of 135 patients with itching or skin changes in the genital area, none showed a reaction to 2% BHA (Marren *et al.* 1992).

In other older studies, positive reactions to BHA were observed in groups of 112 to 2000 patients in about 0.3% to 2% of those tested (3 of 112, 1 of 360, 1 of 485, 2 of 487, 11 of 548, 7 of 926, 8 of 2000) (Hausen *et al.* 1994). Cross-reactions in patch tests against BHA and tert-butylhydroquinone have been described (Larsen 1977; Le-Coz and Schneider 1998; White *et al.* 1984), as have cross-reactions with BHT (Fransway 1991), although these are apparently not obligatory (Baes and van Hecke 1990; Tosti *et al.* 1987). In twelve people sensitized to p-tert-butylphenol-formaldehyde resins, no simultaneous reactions to BHA or BHT were found (Zimerson and Bruze 2002). In some cases, oral provocation with mixtures of BHA and BHT in patients with chronic or recurrent urticaria could trigger a pronounced urticarial reaction (Goodman *et al.* 1990; Juhlin 1981; Roed-Petersen and Hjorth 1976). Two of the patients also reacted to provocation with BHA alone, while 2 control subjects showed no reaction (Goodman *et al.* 1990; Roed-Petersen and Hjorth 1976).

Numerous studies on volunteers have been carried out on the sensitizing effect of preparations containing BHA according to various protocols (CIR 1984). However, due to the low BHA content of 0.01–0.2% and the mixed exposure, the results cannot be used for the evaluation.

According to MAK: overall, clinical experience shows that BHA has at most a very low contact sensitizing potential.

Ref: MAK (2012), CIR (2023; 2024) and references quoted therein

SCCS comment

There is no experimental data from animal studies or New Approach Methods (NAMs) on skin sensitisation potential and potency of BHA. The compilations of CIR and MAK provide clinical evidence that BHA can cause allergic contact dermatitis in humans. The majority of these studies were relatively old. Based on these reports, the SCCS regards the risk of skin sensitisation from exposure to BHA in cosmetics as negligible.

3.4.3 Acute toxicity

3.4.3.1 Acute oral toxicity

Based on information presented in various reports (EFSA, 2011; CIR, 2023; MAK, 2012) and on the ECHA dissemination website, the acute oral toxicity of BHA is low.

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3.4.3.2 Acute dermal toxicity

Based on information presented in CIR, 2023 and on information available at the ECHA dissemination website, the acute dermal toxicity of BHA is low.

3.4.3.3 Acute inhalation toxicity

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SCCS comment on Acute Toxicity:

The available information indicates that BHA is of low acute oral and dermal toxicity.

3.4.4 Repeated dose toxicity

Toxicity after repeated dosing has been investigated in a large number of *in vivo* studies of varying duration (from several days up to 2 years) in different animal species (rat, mouse, rabbit, Guinea pig, Syrian hamster, dogs and monkeys) (Overview given in MAK, 2012). In most of the studies, BHA was administered via food and some of the studies were quite old and not performed according to OECD Test Guidelines. Therefore, studies are summarised in tabular form in Annex I. A more recent oral repeat dose toxicity study focussed on adipogenesis.

Repeated dose oral toxicity

Main target organs after repeated oral dosing of BHA are liver (in rat, mouse, dog and monkeys) and forestomach in rats and hamsters. In rats, tumor promoting effects were described for urinary bladder.

Liver effects:

The most sensitive endpoint in liver was an increase in liver weights, starting at doses higher than 50 mg/kg bw per day. In a long-term study on rats, increased incidences of liver hyperplasia were found in the medium dose groups (230 and 428 mg/kg body weight) in a non-dose-dependent manner (Ito *et al.* 1986 a).

Dog: Increases in liver weight were seen in male and female dogs in a six-month feeding study from as little as 54 mg/kg body weight. The authors explained the liver enlargement with the known proliferation of the endoplasmic reticulum. Apart from a slight reduction in cytoplasmic basophilia, no other pathological changes were observed in the liver cells. No pathological findings were found in the stomach, esophagus, small intestine or liver (Tobe *et al.* 1986). From another study, a dose of 50 mg/kg body weight can be derived for dogs without histopathological effects. In the highest dose group (250 mg BHA/kg body weight), liver cell degeneration and diffuse granulocyte infiltration were observed. The Kupffer cells had deposited increased amounts of hemosiderin, and focal accumulations of the bile pigment were present in the area surrounding the portal vein (Wilder *et al.* 1960). In a one-year feeding study No effects occurred up to a dose of 100 mg/kg body weight. The absolute and relative weights of the liver, kidneys, lungs, brain, heart and spleen were examined. BHA could not be detected in the tissue samples of fat, liver, brain and kidneys or in the urine. The haematological and histopathological examinations of the organs showed no abnormal findings (Hodge *et al.* 1964).

Monkey: In two of 2-3 juvenile monkeys, a non-statistically significant increase in liver weight occurred after 4 weeks of administration of 50 mg BHA/kg body weight (Allen and Engblom 1972; Branen *et al.* 1973). There were no histopathological abnormalities. Only at the next higher dose of 500 mg/kg body weight per day was proliferation of the smooth endoplasmic reticulum evident. It can be assumed that no adverse effects occur in monkeys at a dose of 50 mg/kg body weight. After 85 days of administration, a dose-dependent increase in liver weight was also observed at 125 mg BHA/kg body weight and above (Iverson *et al.* 1985 b, 1986).

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Ref.: MAK (2012) and references quoted therein

Forestomach:

Studies performed in rats and hamsters indicate damage to the forestomach, such as inflammation, hyperkeratosis and hyperplasia, found by Nera *et al.* (1984) after administration of BHA at concentrations of 0.1% or more in the feed (approx. 100 mg/kg body weight) (Altmann *et al.* 1986; Clayson *et al.* 1986; Iverson *et al.* 1985 a; Lam and Garg 1991; Masui *et al.* 1986 a). The effects in the forestomach are assumed to be caused by irritation or local toxicity. However, the forestomach effects are species-specific and cannot be easily transferred to humans.

Ref.: MAK (2012) and references quoted therein

Urinary bladder

See section Carcinogenicity

Adipogenesis

Groups of 24 male C57BL/6J mice each received either normal diet (ND; 10 kcal% from fat) or high fat diet (HFD; 45 kcal% from fat). Each group was further subdivided into three groups, receiving vehicle (0.2% DMSO in physiological saline), 3-BHA (purity > 98%) at 1 mg/kg bw/d or 3-BHA at 10 mg/kg bw/d for 18 weeks. Food consumption was recorded weekly from week 3 to 18 and overnight fasting glucose levels were monitored in weeks 0, 2, 8 and 16. At study termination, blood and tissue samples were taken for haematology, plasma biochemical analysis, histopathological investigation and qPCR. The fat pads of perigonadal visceral white adipose tissue (pWAT) and inguinal subcutaneous white adipose tissue (iWAT) were carefully dissected and weighed to evaluate the fat mass relative to the individual body weight.

Haematological analysis (neutrophils, haematocrit, haemoglobin, lymphocyte, mean corpuscular haemoglobin, mean corpuscular haemoglobin concentration, mean corpuscular volume, lymphomonocytes, mean platelet volume, platelet distribution width, platelets, red blood cells, red blood cell volume distribution width, and white blood cells) did not reveal any differences between exposure groups. All groups of mice on ND had a comparable body weight gain. Control mice of HFD had a higher body weight gain when compared to mice on ND. Compared to HFD controls, HFD mice dosed with 1 mg/kg bw 3-BHA had a lower body weight gain, whereas mice at 10 mg/kg bw 3-BHA had an increased body weight gain. In the ND-fed condition, 3-BHA treatments slightly increased the relative weights of mouse iWAT and pWAT when compared with the DMSO controls. In HFD animals, both iWAT and pWAT were significantly increased. Treatment with 1 mg/kg bw 3-BHA caused slight reduction in relative fat masses of iWAT and pWAT, whereas the administration of 10 mg/kg 3-BHA resulted in a slight increase in iWAT mass and significant weight elevation in pWAT. These alterations were consistent with the mouse body weight gain results. 3-BHA had little impact on glucose metabolism in male mice under the ND or HFD condition.

To assess whether the alterations in adipose tissue accumulation in 3-BHA treatments were related to changes in circulating lipid levels, four parameters of blood lipids—Triglycerides (TG), Total-Cholesterol, HDL-Cholesterol, and LDL-Cholesterol—were investigated. 3-BHA treatments at 1 mg/kg and 10 mg/kg significantly elevated TG levels in both types of mice. Treatments of 1 mg/kg and 10 mg/kg 3-BHA significantly increased total-Cholesterol levels in ND-fed mice and slightly increased total-Cholesterol in 3-BHA-treated HFD-fed mice. Similar findings on HDL-Cholesterol were obtained. Transcriptional levels of the crucial genes involved in adipogenesis, lipid metabolism, and adipocyte endocrine function were further investigated. HFD significantly induced transcriptional levels of PPAR γ , CD36, and Adipo and decreased the mRNA levels of Srebp1c, ACC, and HSL when compared with ND, whereas gene expressions of IL6 and TNF α remained unchanged. Upon the 3-BHA treatment, the mRNA levels of PPAR γ , Srebp1c, CD36, IL6, and TNF α in pWAT were significantly increased in a dose-dependent manner for both ND- and HFD-fed groups. As for ACC and HSL, their transcriptional levels exhibited a dose-related increase in HFD-fed mice with 3-BHA treatment; however, these remained unchanged or slightly decreased in

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ND-fed mice. Regarding the biomarker of Adipo, its mRNA level was not significantly affected by 3-BHA exposure in ND-fed mice. In the HFD group, the Adipo mRNA level was slightly induced by the 1 mg/kg 3-BHA exposure but significantly decreased, owing to the 10 mg/kg 3-BHA treatment. The study authors concluded that their study provided direct evidence of increased adipogenesis and lipid accumulation in both ND- and HFD-fed mice with long-term 3-BHA exposure.

Ref.: Sun *et al.*, 2020

SCCS comment

This is an explorative study indicating that BHA may act as an obesogen. However, the study cannot be used to identify a point of departure for risk assessment.

Repeated dose dermal toxicity

Groups of 10 male and 10 female CD1 mice were epicutaneously administered 0.1 ml of a solution of 30 mg BHA (approx. 1000 mg/kg body weight) in DMSO three times a week for 4 weeks. Repeated dermal dosing did not result in lethality or lung toxicity.

Ref.: Miyakawa *et al.* 1986

SCCS conclusion on repeated dose toxicity

After oral administration, the forestomach and liver were identified as target organs in oral repeat-dose studies in various animal species of different durations (see Annex I). The totality of these studies point to an NOAEL of 50 mg/kg bw/d based on liver effects. Systemic effects were not observed after repeated dermal administration of BHA.

Therefore, the SCCS will use 50 mg/kg bw/d as a point of departure (PoD) for MoS calculation.

3.4.5 Reproductive toxicity

3.4.5.1 Fertility and reproduction toxicity

Fertility and toxicity to reproduction was investigated in a study similar to OECD TG 415. Groups of 12 male and 12 female Sprague-Dawley rats each received 0, 10, 100 or 500 mg BHA/kg bw/d (purity 99.9%) by gavage, dissolved in corn oil, two weeks before mating and for a maximum of two weeks during mating until sperm appeared in the vaginal smear. Male parental (F0) animals were treated for seven weeks, female F0 animals for a period of ten weeks, including three weeks of pregnancy and three weeks of lactation. One to two pups from each litter, a total of 12 offspring per sex and dose group, were treated with BHA from the 21st postnatal day to the age of 13 weeks. In the high dose group (500 mg/kg bw/d), a decrease in mating index and a prolonged mating time was observed in F0 parental animals, whereas body weight was not changed. After 7 weeks of treatment, serum thyroxine (T4) levels decreased significantly at the high dose and testosterone levels decreased significantly in male F0 rats at doses ≥ 100 mg/kg bw/d. In the F0 rats, the absolute and relative organ weights of the liver, adrenal gland and thyroid were increased (in both sexes) and those of the spleen and prostate were decreased. In particular, in female F0 animals, relative liver weight was dose-dependently increased at 100 and 500 mg/kg bw/d reaching statistical significance at the highest dose. In F0 males, relative liver weights were statistically significantly increased at 100 and 500 mg/kg bw/d, whereas the increases in thyroid and adrenal weights reached statistical significance only at the high dose males. Decreases in spleen weights were observed in males and females, reaching statistical significance in high dose males. Relative ventral prostrate weight was statistically significantly decrease in high dose males.

After 13 weeks, the absolute and relative organ weights of liver, adrenal gland and thyroid were increased and those of the spleen, prostate, testis and vagina were statistically significantly decreased in F1 rats at 100 mg/kg bw/d (more than 10% deviation from controls: relative vagina weights in F1 females and of spleen weights in males). The highest dose of

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500 mg/kg/d induced slight dysfunction and underdevelopment of the reproductive system in male and female rats in F1 animals, with a decrease in T4 concentration and testosterone levels, the weights of the sexual organs and a delay in sexual maturity. At the highest dose of 500 mg/kg bw/d, a significant decrease in sperm velocity and motility was observed in F1 rats after 13 weeks. The anogenital distance was slightly shortened on postnatal day 21, but no shortening was detectable after 10 weeks. From the lowest dose of 10 mg/kg bw/d, the width and the ratio between width and length of the sperm head were significantly reduced in a dose-dependent manner.

Ref.: Jeong *et al.* 2005

SCCS comments

The original study report was not available; this information was taken from a publication in the open literature. Deficits of the study consisted in the fact that (I) only a 2-week premating period was used and (II) that food was not controlled for phytoestrogen content. Anses (2024) indicated that total genistein equivalents in the diet used might have exceeded the recommended range. Based on the uncertainties associated with this study, the SCCS does not consider it to derive a point of departure for risk assessment.

3.4.5.2 Developmental Toxicity

Male and female Sprague-Dawley rats were fed 0, 0.125, 0.25 and 0.5% BHA at least 14 days before mating and during mating. For females, feeding continued during pregnancy and lactation, and received as well the same amounts of BHA for up to 90 days after birth. 0.5% BHA in the diet corresponded to 420 mg/kg bw/d day before mating, 410 mg/kg bw/d day during pregnancy and 800 mg/kg bw/d during lactation. At the highest dose, increased mortality and delayed weight gain was observed in the offspring between PND 12 and 30. Food concentrations of 0.125% and 0.25% BHA (corresponding to 220 mg/kg bw/d before mating, 210 mg/kg bw/d during pregnancy, and 440 mg/kg bw/d during lactation) did not have effects on the offspring in terms of growth, reproduction and mortality. The slightly increased but not statistically significant mortality (8.3%) observed with a feed concentration of 0.25% BHA only observed between PND 12-30 was within the range of the control. Changes were not observed in brains or eyes, but other organs were not examined. In behavioral tests before (performed between PND 12-30), a significantly delayed development of the auditory startle response was observed in the offspring at 0.25% BHA in the diet (equivalent to 440 mg/kg body weight and day during lactation). Maternal toxicity was not observed. NOAELs of 210 mg/kg bw/d and 400 mg/kg bw/d were reported for prenatal developmental toxicity and for maternal toxicity, respectively.

Ref.: Vorhees *et al.* 1981

In a study performed according to OECD TG 414, groups of pregnant Wistar rats (group size not specified) received BHA dissolved in sesame oil via gavage doses of 200, 400 or 800 mg/kg bw/d from GD 6 to 15. Controls received vehicle only. In dams of the lowest dose, no changes in body weight gain, food consumption, general symptoms or number of corpora lutea and implantations were observed. At 400 mg/kg bw/d and above, salivation, paralysis of the hind limbs and reduced food consumption were observed in BHA treated animals. In the foetuses, no treatment-related changes in the number of live births, sex ratio, foetal weights or external or internal malformations were observed. Skeletal examination revealed significantly higher incidences of wavy ribs in the fetuses of the mid and high dose when compared to controls.

Ref.: Kawashima *et al.* 1990, only abstract available

Pregnant Sprague-Dawley rats received by oral gavage BHA dissolved in corn oil from GD 7-17 at doses of 250, 500, 750 or 1000 mg/kg bw/d. In dams, a dose-dependent increase in relative liver and stomach weights was observed (no further details). BHA did not lead to any

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anomalies; however, skeletal examination revealed division of the thoracic vertebrae of the fetuses at the highest dose.

Ref.: Han *et al.*, quoted from MAK, 2012.

Diets containing 0 and 0.5% BHA (equivalent to approximately 0 and 750 mg/kg bw/day) were given to mated pairs of Swiss Webster. The litters obtained from the mated pairs were culled to 8 pups at birth and weaned at 21 days of age. The young were then continued on a diet that was similar to that of their mothers. When the pups were 6 weeks old, behavioural testing was initiated. A number of behavioural changes were observed including decreased orientation reflex, decreased sleeping, decreased grooming of self, slow learning and increased exploration.

Ref.: Stokes and Scudder, 1974, quoted from EFSA, 2011

Doses of 0, 50, 200 or 400 mg BHA/kg bw/day were given orally by gavage to groups of pregnant New Zealand white rabbits from day 7 to day 18 of pregnancy. There were no observed differences between the treated groups and controls in the parameters measured (body weight, incidence of soft tissue or skeletal abnormalities, number of foetuses born dead or alive, number of corpora lutea and implantations, general reproductive parameters). The EFSA Panel concluded that a NOAEL for developmental toxicity of 400 mg/kg bw/day can be derived from this study.

Ref.: Hansen and Meyer (1978), quoted from EFSA, 2011

A diet containing the equivalent of 0, 50, 200, or 400 mg BHA/kg bw/day was given to groups of 10 young adult Danish Landrace gilt pigs. The pigs were maintained on the diet for three weeks and then artificially inseminated. Feeding of the diet was continued throughout the study from mating (artificial insemination) to day 110 of the gestation period when the foetuses were removed. No effect was seen on food consumption. A significant effect was observed on the body weight of dams in the dose group given 400 mg/kg bw/day compared to the control group. There was a dose-related increase in absolute and relative liver and thyroid weight and absolute and relative liver and thyroid weight were statistically significantly increased relative to the control at all dose levels tested. There was no effect of BHA on the reproductive and developmental parameters. Anses noted that the histological picture of the thyroid indicated a reduced activity which was not confirmed by other parameters. The EFSA Panel concluded that a NOAEL for developmental toxicity of 400 mg/kg bw/day can be derived from this study and that the NOAEL for maternal toxicity is 200 mg/kg bw/day.

Ref.: Hansen *et al.*, 1982, quoted from EFSA, 2011 and Anses, 2024

The chronic ingestion of 0.5% BHA (corresponding to 750 mg/kg bw/d) by pregnant mice and their offspring resulted in a variety of behavioral changes. Compared to controls, BHA-treated offspring showed increased exploration, decreased sleeping, decreased self-grooming, slower learning, and a decreased orientation reflex.

Ref.: Stokes and Scudder, 1974

Anses (2024) mentions a further, poorly reported study (Clegg, 1965) administering BHA at 750 mg/kg bw/d BHA on GD 1-20 (n=32 females), BHA at 750 mg/kg bw/d for 70 days before pairing, continuing through pregnancy (n= 47 females) and a single administration of BHA on GD 9, 11 or 13 (n=111 females) to Albino rats. BHA also administered at 500 mg/kg bw/d Porton SPF rats for 7 weeks before pairing continuing during gestation; treatment commenced when rats were 3 or 10 weeks old at (n=152 females). BHA was also administered to Imperial Chemical Industries (ICI) SPF mice for 7 weeks before pairing, continuing until GD 18 (n=95 females). In mice, there were no effects on body weight, mortality rates were 2% after two weeks and 25% after 7 weeks. In mice aborting after treatment with BHA there were 74 implantation sites, but only 40 corpora lutea of pregnancy were observed. In Albino rats, weight loss was reported and in Porton SPF rats, growth was impaired.

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SCCS comment on reproductive and developmental toxicity

The reproductive and developmental toxicity of BHA was investigated in a variety of *in vivo* studies in different species. Often, studies were quite old and not performed according to modern standards. Studies indicate effects on fertility (sperm), developmental neurobehavioural effects, and indications for malformations. However, neither study was considered sufficiently robust to justify the selection of a PoD for risk assessment.

3.4.6 Mutagenicity / genotoxicity**3.4.6.1 Mutagenicity / genotoxicity *in vitro***

An overview on *in vitro* mutagenicity/genotoxicity studies performed with BHA is given in **Table 2**.

	Test	Results	SCCS evaluation	Reference
Bacterial gene mutation test (Ames test)				
Bacterial reverse mutation assay	o- and p- tert. BHA (practical grade) from Fluka AG Chemische Fabrik, Switzerland Concentrations: 10, 100 and 1000 µg/plate S. typhimurium strains TA98, TA100, TA1535, TA1537, TA1538 -/+S9	Negative Bacteriotoxicity for all strains observed at 1000 µg/plate.	Negative Limited reliability Only 3 concentrations tested, strong bacteriotoxicity at the highest concentration; 4/5 required strains used	Joner PE, 1977
Bacterial reverse mutation assay	S. typhimurium strains TA98, TA100, TA1535, TA1537, TA1538	Negative	Cited after Environment Canada, Health Canada, July 2010. Publication not accessible to the SCCS.	Bonin and Baker, 1980
Bacterial reverse mutation assay	BHA supplied from the Japan Food Additives Association, Tokyo, at the request of the Ministry of Health and Welfare of Japan, where the purity and quality of each sample were checked. Three concentrations tested up to 300 µg/plate S. typhimurium strains TA92,	Negative	Negative Limited reliability. Only 4/5 required strains used.	Ishidate <i>et al.</i> , 1984

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	TA94, TA98, TA100, TA1535, TA1537 -/+S9			
Bacterial reverse mutation assay	S. typhimurium strains TA98, 100	Negative	Cited after Environment Canada, Health Canada, July 2010. Publication not accessible to the SCCS.	Kawachi <i>et al.</i> , 1980
Bacterial reverse mutation assay	BHA, from Nutritional Biochemicals Corp. (Cleveland, Ohio); purity unknown Concentrations up to 50 µg/plate S. typhimurium strains TA100, TA98 -/+S9	Negative	Negative Limited reliability. 2/5 required strains used.	Shelev and Chin, 1980
Bacterial reverse mutation assay	BHA from Wako Pure Chemical Industries, Ltd., Osaka; purity 99.4% Concentrations: 0.5, 1, 2.5, 5, 10, 25, 50, 100, 250 µg/plate -/+S9 S. typhimurium: TA97, TA98, TA100, TA102 BHQ, BQ also tested.	Negative Bacteriotoxicity for all strains observed at 100 and 250 µg/plate.	Negative Limited reliability. 4/5 required strains used (TA1535 lacking).	Matsuoka <i>et al.</i> , 1990
Bacterial reverse mutation assay	Food-grade BHA: gift from Jan Dekker Company, Wormerveer (The Netherlands); purity unknown S. typhimurium: TA97, TA100, TA102, TA104 -/+S9 Concentrations: 1, 10, 50, 100, 200, 500 and 1000 µg/plate TBHQ from Aldrich.	Negative Bacteriotoxicity for all strains observed at 500 µg/plate, strongest for TA102. No mutagenicity for TBHQ.	Negative Limited reliability 3/5 required strains used (TA1535 + TA98 missing). TA102 and TA104 developed to test oxidative mutagens that are not detected efficiently by other tester strains.	Hagemann <i>et al.</i> , 1988
Bacterial reverse mutation assay	BHA (food grade; 6.6% 2-isomer, 93.2% 3-isomer) from Eastman Kodak (Rochester, NY, USA).	Negative	Negative Limited reliability Only 3 concentrations	Williams <i>et al.</i> , 1990

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	S. typhimurium TA98, 100, 1535, 1537, 1538 -/ +S9 Concentrations: 1, 10, 100 ug/plate		tested. 4/5 strains used (TA102 missing. Bacteriotoxicity observed at 100 µg/plate for TA1537 -S9, and for TA1535 +S9.	
Gene mutations in mammalian cells				
Mammalian cell gene mutation assay	OECD TG 476 (<i>In vitro</i> Mammalian Cell Gene Mutation Test), not GLP compliant CHO cells <u>with</u> metabolic activation (preliminary dose- finding test without metabolic activation) Concentrations: 0, 0.5, 1.0, 2.5 or 5.0 mM.	Negative in the presence and absence of metabolic activation	Study reports not available to the SCCS	Anses (2024), therein quoted as unpublished study reports 2015a and 2015b.
Gene mutation (HGPRT)	BHA (food grade; 6.6% 2-isomer, 93.2% 3-isomer) from Eastman Kodak (Rochester, NY, USA). Adult rat liver (ARL) cells, line 18, at passage 25 Exposure: 3 days Expression: 21-23 days Concentrations: 60, 70, 80, 90 µg/mL - S9	Negative Toxic at 90 µg/mL	Negative Limited reliability S9 was not used, cell enzyme activity not known.	Williams <i>et al.</i> , 1990
Gene mutation (HGPRT) according to own protocol.	BHA, containing approximately 95% 3-tert-butyl- 4-methoxyphenol and 5% 2-tert- butyl-4- methoxyphenol, purchased from Eastman Chemical Products, Inc., Kingsport, TN V79 Chinese hamster lung cells -/ +S9 Concentrations: 18, 36 and 54 µg/mL Metabolic activation by rat or hamster hepatocytes. Exposure: 48 h Expression: 6 days Frequency of 6- thioguanine	Negative Marginal cytotoxicity at 54 µg/mL.	Negative Limited reliability 3 concentrations tested, the highest induced only marginal cytotoxicity.	Rogers <i>et al.</i> , 1985

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	resistant mutants was determined from 20 Petri dishes (60 mm) by counting mutant colonies after 10-12 days.			
Chromosomal aberrations in vitro				
Chromosomal aberrations	BHA from National Institute of Hygienic Sciences, Japan; purity unknown. Pseudo-diploid Chinese hamster cell line (Don) (additional G-band in chromosome 1) -/+S9 Concentrations: 1, 10, 100, 1000 µM	Negative	Negative Limited reliability 100 metaphases scored, only breaks/cell provided in the paper, no detailed data.	Abe and Sasaki, 1977
Chromosomal aberrations	BHA (2(3)-tert.-butyl-4-hydroxyanisole, mixed isomers) from Sigma Chemical Co. Ltd. Purity unknown Chinese hamster ovary cells, clone CHO-K1-BH4 Exposure: for 5 h - S9 and 3 h +S9, Exp I: 33, 100 and 300 µM -/+S9 Exp II: 0, 25, 50, 100, 250, 500 µM +S9. Chromosome analysis: 24-h after the start of treatment. Rates of H2O2 generation were estimated using the peroxidative formation of formaldehyde from methanol in the presence of excess catalase.	Positive +S9 At 500 µM, the mitotic index was reduced to such an extent that chromosome analysis was impossible. No H₂O₂ production when BHA was incubated for prolonged periods in the absence of microsomes. S9 mix under the same conditions produced detectable quantities of H₂O₂ and the rate was increased in the presence of BHA. Maximum stimulation occurred with 250 µM BHA.	Positive +S9 Limited reliability 100 metaphases scored.	Phillips <i>et al.</i> , 1989
Chromosomal aberrations	BHA supplied from the Japan Food Additives Association, Tokyo, at the request of the Ministry of Health and Welfare of Japan (purity	Negative	Negative Limited reliability 100 metaphases scored, w/o S9.	Ishidate <i>et al.</i>, 1984, Ishidate and Odashima, 1977

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	and quality of each sample checked). Purity unknown. Chinese hamster fibroblast cell line -S9 Exposure: 3 concentrations up to 30 µg/mL for 24 and 48 hr			
Chromosomal aberrations	CHO cells	Positive +S9	Cited after Environment Canada, Health Canada, July 2010. Publication not accessible to the SCCS.	Murli and Brusick, 1992
Chromosomal aberrations	BHA from Wako Pure Chemical Industries, Ltd., Osaka; purity 99.4% Chinese hamster lung fibroblasts, CHL cells Treatment -/+S9 for 6 h with harvest 18h later: 0.05, 0.075, 0.1, 0.125, 0.15 mg/mL Treatment -S9 for 24 and 48 h: 0.02, 0.04, 0.06, 0.08 mg/mL	Weakly positive +S9 BHA did not induce chromosomal aberrations at doses up to 0.06 mg/ml (-S9 mix) at which distinct cytotoxicity toxicity was noted. With S9, BHA induced chromosomal aberrations with low frequency (11%) at 0.125 mg/ml. BHQ and BQ, showed similar results: 19% at 0.04 mg/ml and 11% at 0.02 mg/ml, presence of S9 mix. The order of clastogenic potential in the presence of S9 mix was BQ>BHQ>BHA. These findings suggest that some metabolites other than BQ and BHQ are responsible for the induction of chromosomal aberrations by BHA.	Positive +S9 Limited reliability 100 metaphases scored.	Matsuoka <i>et al.</i> , 1990.
Other endpoints				
Comet assay	BHA from Sigma, purity unknown A549 lung carcinoma cells Comet assay in alkaline conditions: 24 h Concentration: 1 mM Image analysis by CASP software	Positive %DNA in tail several fold higher than in control, and almost as high as positive control H ₂ O ₂ 200 µM. However, In MTT test: 100% growth inhibition at 1 mM In apoptosis assessment: >54% of BHA-treated cells showed the late stage of apoptosis (Annexin V-positive/ PI-positive staining), and 33.40% of the cells were in the early stage of apoptosis (Annexin V-positive/PI-negative) after 24 h.	Study not reliable Considering that MTT test shows 100% growth inhibition, and PI/Annexin V staining shows 54.21% in late apoptosis stage, there is strong evidence that BHA at the only one concentration used (1mM) after 24 h induced severe cell death, so the comet results are not reliable.	Vandghanooni <i>et al.</i> , 2013

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Comet assay	BHA from Sigma, purity unknown Human peripheral blood lymphocytes and sperm cells Comet assay in alkaline conditions: 0.5 h Concentration: 0.5 mM Manual scoring	Positive BHA <i>per se</i> produced a pro-oxidant response in both lymphocytes and sperm (at least 2-fold increase in DNA damage comparing to control). The presence of BHA reduced the levels of DNA damage in lymphocytes in all oestrogenic treatments and it abolished strand breaks against H ₂ O ₂ .	Positive Limited reliability. Short incubation time, details on methodology not provided. 50 cells from each duplicate scored per concentration per experiment. Olive tail moment used as the DNA damage endpoint.	Cemeli <i>et al.</i> , 2011
Sister chromatid exchanges according to own protocol.	BHA, containing approximately 95% 3-tert-butyl-4-methoxyphenol and 5% 2-tert-butyl-4-methoxyphenol, purchased from Eastman Chemical Products, Inc., Kingsport, TN V79 Chinese hamster lung cells Concentrations: 18, 36 and 54 µg/mL Metabolic activation was provided by rat or hamster hepatocytes freshly prepared by perfusion of the liver in situ with collagenase. Exposure: 24 h	Negative Marginal cytotoxicity at 54 µg/mL.	Negative Limited relevance	Rogers <i>et al.</i> , 1985
Unscheduled DNA synthesis	BHA (food grade; 6.6% 2-isomer, 93.2% 3-isomer) from Eastman Kodak (Rochester, NY, USA). Hepatocytes isolated from F344rats Exposure: 18-20 h Concentrations: 0.01, 0.1, 0.5, 1, 5 µg/mL - S9	Negative Toxic at 5 µg/mL	Negative Limited reliability S9 was not used, cell enzyme activity not known.	Williams <i>et al.</i> , 1990
Sister chromatid exchange test	BHA (food grade; 6.6% 2-isomer, 93.2% 3-isomer) from Eastman Kodak (Rochester, NY, USA). Chinese hamster ovary (CHO) cells Exposure: 24 h Concentrations: 5, 50, 500 µg/mL -/+ S9	Negative Toxic at 500 µg/mL	Negative Limited reliability. 2 analysable concentrations used.	Williams <i>et al.</i> , 1990

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Sister chromatid exchange test	BHA from National Institute of Hygienic Sciences, Japan; purity unknown Pseudo-diploid Chinese hamster cell line (Don) (additional G-band in chromosome 1) -/+S9 Concentrations: 1, 10, 100, 1000 µM	Negative The t-test for SCE indicated significant increase at 5% level, compared with the mean value shown for solvent.	Equivocal Limited relevance.	Abe and Sasaki, 1977
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The *in vitro* genotoxicity of BHA was investigated in a variety of genotoxicity tests and in indicator tests for genotoxicity. Not all of the tests were performed according to modern standards/guidelines. The available bacterial gene mutation tests covering all 5 strains required by OECD TG repeatedly resulted in negative results. Two mammalian *in vitro* gene mutation studies of limited reliability performed in adult rat liver and V79 cells were negative as well. However, three out of five available *in vitro* chromosomal aberration studies – although of limited reliability – were positive in the presence of S9 mix. Amongst the indicator studies performed, one Comet assay of limited reliability reported positive results.

3.4.6.2 Mutagenicity / genotoxicity *in vivo*

An overview on *in vitro* mutagenicity/genotoxicity studies performed with BHA is given in **Table 3**.

Test		Results	SCCS Evaluation	Reference
Comet assay on liver and kidney cells	BHA from Sigma-Aldrich, purity unknown Rat, Sprague-Dawley (f, n=10), 0.7 mg/kg bw/d for 60 days. Test performed in liver and kidneys. Blood samples collected after 30 and 60 days.	Negative	Not reliable Oral administration not specified. Administered dose eventually too low. Time of tissue sampling not specified. Potential short-lived DNA damage eventually not be detected. Poor description. Method for scoring the slides is not reliable. No data on cytotoxicity in hepatocytes/nephrocytes. No data on historical control values.	Abo-El-Sooud <i>et al.</i> , 2018
Comet assay	BHA from Tokyo Kasei Kogyo Industry Ltd., Tokyo, Japan; purity >98% Male ddY mice (n=4) Doses: single oral at 100, 500 and 1000 mg/kg Sampling after 3 h (all doses) or 24 h (1000 mg/kg).	Positive Increased DNA migration in: - glandular stomach: at 1000 mg/kg after 3 h and after 1000 mg/kg after 24 h - colon: at 500 and 1000 mg/kg after 3 h and after 1000 mg/kg after 24 h Necropsy and histopathological examination of tissue sections stained by HE and TUNEL methods: no effect of	Positive In glandular stomach and colon. Four animals per group were used (5 is required according to OECD TG 489); single treatment (daily treatments over a duration of 2 or more days are required according to OECD TG 489)	Sasaki <i>et al.</i> , 2002

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	Organs examined: stomach, colon, liver, kidney, bladder, lung, brain, bone marrow.	treatment on any organ examined. DNA damage observed not likely to be due to general cytotoxicity (necrosis) and apoptosis.	Study of limited reliability.	
Alkaline elution assay	3-BHA (purity 99.4%) from Wako Pure Chemical Co. Ind. (Osaka, Japan). F344/DuCrj rats Exposure: single oral (gavage) at 220 mg/kg Sampling after 3 hr Organs examined: forestomach mucosa	Negative No detectable DNA damage	Negative Group size not given	Morimoto <i>et al.</i> , 1991
UDS on forestomach epithelium cells	BHA from Sigma, purity unknown B6C3F1 mice (n=5), single oral administration (300 mg/kg). Sampling of stomach after 2 or 4 hr. [³ H]thymidine incubation and autoradiography. Grains per nucleus assessed in 50 cells from each of three preparations from each animal.	Negative	Negative Limited relevance	Benford <i>et al.</i> , 1994
Covalent binding to DNA, RNA, proteins	Radiolabelled 3-BHA and 2-BHA (radiochemical purity 97.5%) from New England Nuclear (Boston, USA). Unlabelled 3-BHA and 2-BHA (> 99% purity) synthesized in Sumitomo Chemical Co., Ltd., Osaka, Japan F344 rat (n=5) single intragastric administration of 1000 mg/kg with or without pretreatment with unlabelled 1% 3-BHA in the diet for 6 days. Sampling (forestomach and glandular stomach, liver, kidney) after 6 hr.	Negative 3-BHA, 2-BHA did not bind covalently to DNA or RNA in the forestomach, glandular stomach or kidney, but in the liver both compounds bound to RNA, their extents of binding being similar. The amounts of radioactivity of these compounds bound to proteins in the 4 tissues were similar.	Negative Limited relevance	Hirose <i>et al.</i> , 1987

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DNA-adduct formation in forestomach cells	3-BHA (>99% purity) synthesized by Sumitomo Chemical Co. Ltd. (Osaka, Japan). Assessment of DNA adducts in forestomach of male F344 rats given either a single or repeated oral gavage administration (5 days) of 3-BHA (1000 mg/kg), tert-butyl-1,4-benzoquinone (TBQ) (125 mg/kg) or 5-methoxy-3-tert-butyl-1,2-benzoquinone (3-TBOQ) (250 mg/kg).	Negative No DNA adducts detected after exposure to BHA, TBQ or 3-TBOQ	Negative Limited relevance	Saito <i>et al.</i> , 1989
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Five *in vivo* studies from the open literature were available to evaluate genotoxicity of BHA. The studies were not performed according to OECD guidelines. Most of the studies were of limited reliability and negative. An *in vivo* Comet Assay considered of limited reliability by the SCCS, increased DNA migration was determined in glandular stomach and colon tissue from mice (in glandular stomach at 1000 mg/kg, in colon starting at 500 mg/kg).

SCCS comment

Based on the observations from *in vitro* and *in vivo* studies performed with BHA, there is a concern with respect to a clastogenic potential. However, the fact that positive findings were observed in *in vitro* chromosomal aberration tests and in an *in vivo* comet assay has to be put into perspective. Positive *in vitro* findings were only observed in the presence of S9, indicating that metabolism might be required to exert clastogenicity. It is known from literature that BHA and its metabolites (e.g. tert-butylhydroquinone) can lead to the formation of reactive oxygen species in the presence of liver microsomes (Cummings and Prough, 1983; Phillips *et al.* 1989) or that BHA could be oxidised to produce free radicals. This is underlined by mechanistic investigations showing that chromosomal aberration was reduced in the presence of catalase (Philipps *et al.*, 1989). As described for other antioxidants, in living organisms, the occurrence of clastogenicity will be dependent on (i) metabolism, (II) exhaustion of antioxidant capacity and, last but not least, on the amounts (systemically) available at the target site.

With respect to the positive findings observed in an *in vivo* comet assay performed in mice, it is of note that positive findings were observed at high doses (500 and 1000 mg/kg). Such doses might indeed have exhausted antioxidant capacities in the target cells, but more importantly, when looking at the toxicological effects of BHA from a more holistic point of view, it should be kept in mind that liver toxicity (after repeated administration) already starts at much lower doses (see section 3.4.4 Repeated dose toxicity).

Therefore, it is most likely that a threshold mode of action can be assumed for the clastogenic effects observed with BHA *in vitro*. Therefore, the SCCS will derive a safety calculation (MoS) based on a PoD resulting from repeat dose toxicity studies (50 mg /kg bw/day).

3.4.7 Carcinogenicity

From Anses, 2024:

Several international and European assessments have been carried out on the toxicological effects of BHA. No study following regulatory guideline is available. As for the other endpoints, the composition of BHA was not always indicated.

Historic

- IARC reviewed BHA in October 1985 and classified it in Group 2B (Possible Human Carcinogen) based on "sufficient evidence of carcinogenicity in experimental animals" and "no data were available on the carcinogenicity" in human beings (IARC, 1987). In a workshop, the mechanism by which BHA induces forestomach tumors in rats and was discussed and it was determined that the mechanism was not relevant to humans (IARC, 2003).

- California Proposition 65: BHA listed on January 1, 1990 as a chemical known to the State to cause cancer under the Safe Drinking Water and Toxic Enforcement Act of 1986 (this decision was based on IARC). A No Significant Risk Level of 4 mg/day was established (OEHHA, 1990).

- Netherlands environmental and public health institute, 2004: BHA is in the RIVM list of 514 potential CMR substances that are not on Annex 1 of 67/548/EEC (RIVM, 2004).

- NTP Report on Carcinogens (RoC): BHA was first listed in the Sixth Annual RoC in 1991 (NTP, 2016)

- NTP, 2016: 14th Edition: "reasonably anticipated to be a human carcinogen" based on sufficient evidence of carcinogenicity from studies in experimental animals. 14th NTP RoC acknowledged that the epidemiological study of stomach cancer in men and women within the Netherlands showed no increase in risk at typical levels of dietary intake of BHA (Botterweck *et al.*, 2000).

Data

Regarding human data, the association between dietary intake of BHA and stomach cancer risk was investigated in the Netherlands Cohort Study (NLCS) that started in 1986 among 120,852 men and women aged 55 to 69 years. A semi-quantitative food frequency questionnaire was used to assess food consumption. Information on BHA content of cooking fats, oils, mayonnaise and other creamy salad dressings and dried soups was obtained by chemical analysis, a Dutch database of food additives (ALBA) and the Dutch Compendium of Foods and Diet Products. After 6.3 years of follow-up, complete data on BHA intake of 192 incident stomach cancer cases and 2035 subcohort members were available for case cohort analysis. Mean intake of BHA among subcohort members was 105 mg/day. For consumption of mayonnaise and other creamy salad dressings with BHA, no association with stomach cancer risk was observed. A statistically non-significant decrease in stomach cancer risk was observed with increasing BHA intake [rate ratio (RR) highest/lowest intake of BHA = 0.57 (95% confidence interval (CI): 0.25- 1.30]. In this study, no significant association with stomach cancer risk was found for usual intake of low levels of BHA (Botterweck *et al.*, 2000).

Regarding animal data, numerous subchronic and chronic exposure studies of BHA administered via food are available. The majority of these studies were conducted in rodents and are prior to 2011, therefore already taken into account in pre-existing evaluations.

Many reviews of these studies are available in the literature: EFSA, 2011; JECFA, 1986; 1989; SCF, 1989; IARC, 2003, NTP, 2016, Felter *et al.*, 2021.

Overall, in rats, BHA was tested at dose levels of 0.125%-2% (55 - 1000 mg/kg bw/day). At dietary concentration of 0.125% or higher, BHA led to significant incidences of forestomach hyperplasia, at dietary concentration of 1% or higher significant incidences of papillomas and at dietary concentration of 2% significant incidences of squamous cell carcinomas. Reversibility of the hyperplastic lesions was shown to be dependent on dose and duration. In a strain comparison study, the SHR rat strain was shown to be most sensitive, with the highest incidence of squamous cell carcinoma. The sensitivity of the different strains was established as follow: F344 rats, 26.7% (8/30); SHR rats, 76.7% (23/30); SD rats, 36.7% (11/30) and Lewis rats, 6.7% (2/30) (Tamano *et al.*, 1998).

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Studies performed in hamsters showed proliferative changes in the forestomach including induction of papillomas at a dose level of 1% BHA (1200 mg/kg bw/day). A study by Hirose *et al.* 1990 was identified in the literature search in which it was shown that squamous cell dysplasia was not reverted at a dose level of 2% BHA.

It was suggested that the tumorigenic action of crude BHA (the authors refer either to the mixture of approximately 98% 3-BHA and 1% or 2% 2-BHA, or to the commercial mixture which is a mixture of 75 to 98% 3-tert-BHA (3-BHA) and 1 to 25% 2-tert-BHA (2-BHA)) on the hamster forestomach was largely due to 3-BHA (Ito *et al.* 1986, Ito *et al.* 1987, Hirose *et al.*, 1986).

Studies performed in mice showed proliferative changes in the forestomach including induction of papillomas and carcinomas at 0.5% BHA (750 mg/kg bw/day).

In species without a forestomach, no histopathological effects were observed. Overall, the studies have included species with and without a forestomach (rats, mice, hamsters, guinea pigs and dogs). Gastric epithelial hyperplasias, papillomas and carcinomas were seen in the forestomach of rodents (EFSA, 2011). No tumours were observed in other tissues and organs in mammals.

The absence of a homologue tissue for the forestomach in humans makes it difficult to extrapolate these observations to humans. The question of the relevance of this tumour type for human hazard has been extensively discussed in the early 2000's. It was noted that if humans do not have a forestomach, they possess histologically similar organs, including the oral cavity, pharynx, oesophagus, and glandular stomach. The potential risk from non-genotoxic forestomach carcinogens for humans involves exposure of the mouth, pharynx, and oesophagus at dose levels that exert irritating action.

It is assumed that exposure to non-genotoxic chemicals at concentrations far below those having irritating potential is not hazardous to humans (Wester *et al.*, 1988). IARC (2003) discussed that non-DNA reactive agents appear to cause forestomach tumours primarily through initial cytotoxicity and subsequent sustained cell proliferation and hyperplasia.

IARC (2003) also recognises that the relevance for humans is probably limited for agents that have no demonstrable genotoxicity and that are solely carcinogenic for the forestomach squamous epithelium in rodents after oral administration. This is because the experimental exposure conditions during oral administration in animals (particularly with gavage dosing) are quite different compared to human exposure. In experimental animals, high local exposure concentrations of test substance in the forestomach and prolonged exposure of the epithelial tissue may have physical effects. Consequently, for these agents, the carcinogenic mode of action might be specific to experimental animals. IARC therefore recommends that, in evaluating the relevance of the induction of forestomach tumours in rodents for human cancer, the exposure conditions in the experiments have to be considered (IARC 2003, shared by RIVM, 2004). In this review, IARC also looked at some substances in particular. Indeed, they concluded that BHA induced only squamous-cell tumours of the forestomach in three rodent species. At high concentrations, the mode of action by which BHA produces forestomach tumours may involve a net production of free radicals during the long transit time in the forestomach, resulting in cytotoxicity and subsequent hyperplasia. Upon chronic administration of BHA, this hyperplasia is sustained, which leads to a tumour-promoting effect specific to the forestomach.

In more recent literature, it was shown that such chemicals should be evaluated in detail before discarding continuous induction of cell proliferation, hyperplasia, and ultimately carcinomas. The genotoxicity studies indicate a lack of potential for BHA to induce mutations or interaction with or damage DNA directly, the effects observed being secondary to oxidative stress.

Overall, BHA induces proliferative changes in the forestomach of rodents. Other species studies, including rats, pigs and indicate that high dose of BHA can target tissues in the digestive system other than the forestomach without leading to tumor.

In conclusion, BHA induced tumours in the forestomach of 3 rodent species (rats, mice and hamster) led IARC to classify BHA as Possible Human Carcinogen in 1986. However, in 2003, IARC recognised that the relevance for humans is probably limited and that exposure relevance should be considered. In addition, no significant association with stomach cancer

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risk was found for usual intake of low levels of BHA in the Netherlands Cohort Study. Concordance is seen between the target tissues in the different species tested (forestomach in rodents, oesophagus in pigs and monkeys) showing that the substance could potentially target particular cell types from the gastrointestinal tract e.g. squamous epithelial cells. Therefore, effects seen in the digestive tract that could be due to epithelial cytotoxicity (i.e. equivalent to corrosion/irritation of the substance at the site of contact).

Ref.: Anses (2024) and references quoted therein

SCCS overall comment on carcinogenicity

The SCCS recognises that BHA induced forestomach tumours in 3 different rodent species. This tumour type is considered not relevant for humans. From a mechanistic point of view, the available information indicated that tumour formation might be due to formation of radicals and/or reactive oxygen species at high doses causing local cytotoxicity and finally leading to tumours. Based on the anticipated low intended use concentration in cosmetic, carcinogenicity is not considered as concern with respect to cosmetic uses.

3.4.8 Photo-induced toxicity**3.4.8.1 Phototoxicity / photo-irritation and photosensitisation**

/

3.4.8.2 Photomutagenicity / photoclastogenicity

/

3.4.9 Human dataToxicokinetics:

Taken from MAK (2012)

After oral administration of 100 mg BHA, maximum plasma levels of 600 ng/ml were measured in one person after two hours and 220 ng/ml in a second person after half an hour (El-Rashidy and Niazi 1979).

Nine male subjects received a dose of 0.5 mg BHA (purity >99%)/kg bw, dissolved in corn oil in a gelatin capsule. The concentrations of free BHA in plasma (Verhagen *et al.* 1987), free and conjugated BHA and tert-butylhydroquinone (TBHQ) in urine and BHA in feces and fatty tissue (Verhagen *et al.* 1989 b) were determined. The highest concentration in plasma was $117 \pm 75 \mu\text{g}$ free BHA/l after about 0.7 hours (value from figure). Oral absorption, measured as excretion of BHA or its conjugates in urine, was between 32 and 77% (Astill *et al.* 1962), 50% (Verhagen *et al.* 1989 b) and 100% (ElRashidy and Niazi 1983).

In four subjects, a half-life of 3 hours was determined after administration of 5 or 30 mg 3-tert-butylhydroxyanisole (Castelli *et al.* 1984). In nine subjects, the half-life of free BHA in plasma after administration of 0.5 mg BHA/kg body weight was 58 ± 33 minutes and the AUC was $191 \pm 51 \mu\text{g/l.h}$ (Verhagen *et al.* 1989 b).

After oral administration of 0.5 to 1.7 mg/kg body weight, more than half of the dose was excreted in the urine within 24 hours, and TBHQ conjugates and 4-O-glucuronide were also measured in the urine (Castelli *et al.* 1984; Daniel *et al.* 1967; El-Rashidy and Niazi 1983). BHA could not be detected in the faeces (Verhagen *et al.* 1989 b).

Ref.: MAK (2012) and references quoted therein

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Repeated dose toxicity:

Eight male volunteers received 0.5 mg BHA/kg bw/d orally for 10 days. Blood samples were taken one day before and on the 4th, 6th and 7th day after the first dose to determine the activities of L-aspartate aminotransferase, L-alanine aminotransferase, L-gammaglutamyltranspeptidase, creatine phosphokinase, lactate dehydrogenase and the levels of total protein, albumin, urea, creatinine, sodium and chloride. BHA did not cause effects on clinical chemistry parameters or on phase I and phase II biotransformation capacity measured with antipyrine and paracetamol.

Ref.: Verhagen *et al.* 1989 c

3.4.10 Special investigations

Endocrine Disruption

Level 1: in *silico* data

The SCCS *in silico* analysis used various models. All models resulted in a negative outcome:

Estrogen Receptor-mediated effect (IRFMN-CERAPP) 1.0.1

This model relates not only to binding but includes further steps along the AOP. Not active. Prediction with good reliability. A very similar (isomer) substance is not active. Other similar substances not active.

Estrogen Receptor Relative Binding Affinity model (IRFMN) 1.0.2

This model only relates to the binding to the nuclear receptor. Prediction Active. Reliability moderate. Indeed, the most similar substance, an isomer, is not active.

Androgen Receptor-mediated effect (IRFMN-COMPARA) 1.0.1

Not active. This model relates not only to binding but includes further steps along the AOP. Prediction with good reliability. A very similar (isomer) substance is not active. Other similar substances not active.

Thyroid Receptor Alpha effect (NRMEA) 1.0.1

Not active. This model only relates to the binding to the nuclear receptor. Prediction with good reliability. A very similar (isomer) substance is not active. Other similar substances not active.

Thyroid Receptor Beta effect (NRMEA) 1.0.1

Not active. This model only relates to the binding to the nuclear receptor. Prediction with good reliability. A very similar (isomer) substance is not active. Other similar substances not active.

Glucocorticoid Receptor (OBERON) 1.0.0

This model only relates to the binding to the nuclear receptor.

Not active. Prediction with good reliability. A very similar (isomer) substance is not active. Other similar substances not active.

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Thyroperoxidase Inhibitory Activity (OBERON) 1.0.1

Not active. Prediction with low reliability. Indeed, using read-across a very similar (isomer) substance is active. Other similar substances active.

Steroidogenesis (OBERON) 1.0.0

Not active. Prediction with good reliability. A very similar (isomer) substance is not active. Other similar substances not active.

Level 2: *in vitro* mechanistic data

Taken from Anses (2024)

A) ToxCast: Models and Computational Toxicology bioactivity data

The following table gathers the output of Toxcast: Models based on *in vitro* tests for 2-tertButyl-4-methoxyphenol (CAS 121-00-6).

[2-tert-Butyl-4-methoxyphenol - Chemical Details \(epa.gov\)](https://www.epa.gov/chemical-data/2-tert-butyl-4-methoxyphenol-chemical-details)

Model	Receptor	Agonist	Antagonist	Binding
ToxCast Pathway Model (AUC)	Androgen	0	0	-
ToxCast Pathway Model (AUC)	Estrogen	0.0139	0	-

AUC: area under the curve; Model scores range: 0<AUC<1 (bioactivity of 17 β -estradiol); AUC=0: no activity; 0.001<AUC<0.1: inconclusive bioactivity; AUC $\geq$ 0.1: positive bioactivity; -: not tested.

The ToxCast Models shows that 3-BHA is inactive for AR agonist and antagonist modes and ER antagonist mode. 3-BHA is an inconclusive ER agonist. Those models predict that 3BHA should not bind to the androgen receptor or have androgen activity. The inconclusive ToxCast estrogen receptor pathway model result suggests that estrogen activity *in vitro* may be at concentrations that approach the range of cytotoxic concentrations *in vitro*.

In addition, regarding the thyroid modality, the CCTE_Simmons_AUR_TPO assay appeared positive in Comptox [v2.4.1](#) indicating 3-BHA may act on thyroperoxidase (TPO) activity. The NVS_NR_hTRa (thyroid hormone receptor, alpha) Antagonist assay and the NVS_TR_hDAT assay (related to the dopamine transporter SLC63 (or DAT) gene) were also active.

B) experimental studies

Level 2 *in vitro* mechanistic data on endocrine activity of BHA are compiled in table XX below

Table 4: Level 2 *in vitro* mechanistic data on endocrine activity of BHA (compiled from Anses, 2024; references quoted therein)

Test/Testsystem	Results	Conclusion	Reference
Estrogenic Pathway MCF-7 cell proliferation assay (E-Screen assay)	Induction of MCF-7 cell proliferation (lowest concentration inducing maximal proliferation: 50 μ M; relative proliferative	Weak estrogenic activity	Soto <i>et al.</i> (1995)

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MCF-7 cell proliferation assay (E-Screen assay)	effect of 30% compared to estradiol Induction of cell proliferation, maximal proliferation at 50 µM; inhibition of cell proliferation by anti-estrogen ICI 182,780; 200 and 900 µM of BHA inhibited 50% of 17β-estradiol binding to ERα and ERβ. 50 µM BHA decreased ERα- gene expression and time-dependently increased progesterone receptor gene expression	BHA can bind to ERα and ERβ	Okubo and Kano, 2003
ERα and ERβ transactivation assays in U2-OS ERα and U2OS ERβ cell lines	BHA-induced induced maximal transcriptional activities: 18.3% and 15.6%, for ERα and ERβ respectively, compared to estradiol-induced maximal activities.	Estrogenic activity of BHA; potency for ERα lower than for ERβ mediated estrogenic activity.	Ter Veld <i>et al.</i> (2006)
Transcriptional activity in breast cancer ZR-75 cells and MCF-7 cells.	Stimulation of estrogen receptor in ZR-75 cells, transcriptional activity in MCF-7 cells.	Weak estrogenic activity of BHA.	Jobling <i>et al.</i> (1995)
Estrogenic and anti-estrogenic activities of BHA in T47DKbluc cells.	Uncertainties in viability tests.	Conclusion not possible.	Pop <i>et al.</i> (2014)
Estrogenic and anti-estrogenic activities of BHA in T47D-Kbluc and MCF-7 cells-	Cell proliferation in MCF-7 cells; anti-estrogenic effects in T47D-Kbluc cells investigated by incubation of BHA in presence of 30 pM E2	Weak estrogenic activity of BHA; anti-estrogenic activity of BHA.	Pop <i>et al.</i> (2018)
Estrogenic activity of BHA in MVLN luciferase activity assay	Exposure to BHA (0-10 µM) did not significantly alter luciferase activities in MVLN cells. Furthermore, no ER binding affinity was observed in a Hithunter ER binding assay.	BHA not estrogenic.	Yang <i>et al.</i> (2018a)
Androgenic Pathway Investigation of AR transactivation in PC-3 LUC AR+ cells; BHA effect on dihydroxytestosterone (DHT) response.	Transactivation assay negative; BHA-induced, partial inhibition of DHT response.	Weak anti-androgenic activity of BHA.	Schrader and Cooke, 2000
AR transactivation in MDA-kb2 cells.	In presence of 0.5 nM DHT, 50 µM BHA induced 0.68-fold luciferase activity (compared to solvent control).	Weak anti-androgenic activity of BHA.	Klopčič and Dolenc, 2017
Inhibition of DHT-induced response in MDA-kb2 cells.	Statistically significant, dose dependent anti-androgenic activity of BHA.	Weak anti-androgenic activity of BHA.	Pop <i>et al.</i> , 2016
AR transactivation in MDA-kb2 cells.	Concentration-dependent decrease of DHT-induced luciferase activity by 3-BHA.	Weak anti-androgenic activity of BHA.	Wang <i>et al.</i> , 2023
Effects of BHA on cells expressing 3α-hydroxysteroid dehydrogenase (AKR1C14) and 5-α-reductase (SRD5A1).	BHA (containing minimum 90% 3-isomer and 9% 2-isomer) inhibited SRD5A1 and AKR1C14 activities and weakly inhibited retinol	Antiandrogenic and androgenic activities of BHA.	Guo <i>et al.</i> , 2017

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Activities of HSD11B1 (cortisone -> cortisol) and HSD11B2 (cortisol -> cortisone) microsomes from of rat liver, rat and kidney and human placenta in the absence or the presence of 1 to 100 µM BHA. Incubation of rat Leydig cells in basal conditions or in LH- or 8Br-cAMP-stimulated conditions, with or without various steroidogenic substrates and in the absence (control), or presence of 0.05, 0.5, 5 or 50 µM BHA.	dehydrogenase-2 activity. BHA did not change the activity of HSD11B1, but inhibited that of HSD11B2. BHA (tested only with 50 µM) reduced secretion of testosterone and/or 5α-androstane-diol when cells were cultured in the presence of LH, 8Br-cAMP, pregnenolone, progesterone, androstenedione or testosterone. Androgens secreted were unchanged by BHA in basal condition and in the presence of 22R hydroxy-cholesterol or DHT. mRNA levels of Hsd17b3 and Srd5a1 were significantly decreased by 50 µM BHA whereas those of Scarb1, Star, Cyp11a1, Hsd3b1, Cyp17a1 and Akr1c14 were unchanged. BHA-induced reduction in HSD17B3 and SRD5A1 protein levels BHA-induced reduction of enzymatic activities of HSD3B1 and CYP17A1, but not HSD17B3 in Leydig cell microsomes.	BHA can alter the metabolism of corticosteroids by enhancing the concentration of cortisol compared with that of cortisone (inactive metabolite of cortisol). BHA is an inhibitor of androgen production in rat Leydig cells via direct and indirect inhibition.	Li <i>et al.</i>, 2016a Li <i>et al.</i>, 2016b
Thyroid Pathway (Anti)agonist activity of 3-BHA on Retinoic acid receptor RARα in MCF-7 cells	Antagonistic effects addressed by investigation of 3-BHA induced inhibitory effects on AM580-induced firefly luciferase activity in transfected MCF-7 cells.	Slight antagonistic effects on RARα.	Xu <i>et al.</i>, 2022
3-BHA-induced inhibition of iodotyrosine deiodinase (iodide recycling enzyme) <i>in vitro</i>.	3-BHA tested at a maximum concentration of 200 µM.	Negative	Olker <i>et al.</i>, 2021
Interaction of chemicals with thyroid hormone receptors using the GH3-TRE-Luc cell line.	3-BHA tested at a maximum concentration of 200 µM.	Negative (BHA neither TR agonist nor TR antagonist.	Paul-Friedman <i>et al.</i>, 2019.
Susceptibility of iodothyronine deiodinase enzymes (types 1, 2 and 3) to chemical inhibition.	3-BHA tested at a maximum concentration of 200 µM.	Negative	Olker <i>et al.</i>, 2019

Level 2 *in vitro* mechanistic data indicate estrogenic, steroidogenic activity and anti-androgenic activity of BHA.

Level 3: *in vivo* mechanistic data

10 immature Wistar female rats, aged 17-21 days-old, were exposed to 300 mg/kg bw/day BHA for three consecutive days by oral suspension. Positive control animals were exposed to 17 β -estradiol by subcutaneous injection into dorsal surface at a dose of 20 μ g/kg in sunflower oil. Negative control animals were given the vehicle by oral suspension.

Except for negative and positive controls, no significant changes in body weight were observed in treated groups. The relative liver, kidneys and spleen weights were not affected by treatment with BHA. Compared to the negative control group, relative uterine and ovaries weights were significantly decreased. Rats had minimal histopathological changes in the uterus, with no changes in the cervix and vagina.

Ref.: Pop *et al.*, 2013, Taken from Anses, 2024

SCCS remark on Pop *et al.*, 2013

In line with Anses (2024), the SCCS considers this study not reliable. It is of poor quality (Klimisch 3) and there were deficiencies in methodology and reporting. Therefore, the study is not used to assess the endocrine activity of BHA.

Uterotrophic assay:

In a study from the open literature, (anti-) estrogenic properties of BHA (2(3)-tert-butyl-4-hydroxyanisole [25013-16-5], 99.9%) were investigated in the immature uterotrophic assay. 17 β -estradiol as positive control was used at final dosages of dosage of 2 and 20 μ g/kg bw in corn oil. BHA doses of 50, 100, 250 and 500 mg/kg bw/d were administered once daily subcutaneously for three consecutive days to 20 d old Crj:CD (SD) rats. To investigate anti-estrogenic activity, 50 and 500 mg/kg bw BHA was injected subcutaneously to 20 d old animals, followed by 17 β -estradiol at 2 μ g/kg bd s.c.. Injections were made on three consecutive days. On the third day, rats were sacrificed and weights of uteri were determined with fluid and/or without fluid. For the measurement of uterine epithelial cell height, uterus was fixed to Y-shaped form on formalin pre-wetted filter paper. Each uterine was divided into three parts, representing the proximal, medial and distal part.

Body weight gain was statistically significantly decreased at 250 and 500 mg/kg bw BHA alone. Relative and absolute uterine weights containing fluid were significantly, but not dose-dependently decreased by all doses (50, 100, 250 and 500 mg/kg bw) of BHA alone. The highest dose of 500 mg/kg bw BHA significantly decreased 17 β -estradiol-stimulated increase of relative and absolute uterine and vaginal weight. Uterine epithelial cell height was not changed by treatment with BHA alone. Also, BHA at 50 or 500 mg/kg bw did not affect 17 β -estradiol induced increase of uterine epithelial cell height.

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Table 5

Table 1

Body weight gain, relative liver weight, and serum estradiol after the treatment of BHA or 17 β -estradiol in immature female rats

Groups	Body weight gain (g)	Liver (g/100 g bw)	Estradiol (pg/ml)
Vehicle (corn oil)	8.6 $\pm$ 1.3	4.34 $\pm$ 0.38	14.3 $\pm$ 10.7
17 β -estradiol (20 μ g/kg)	7.0 $\pm$ 1.3	4.28 $\pm$ 0.31	37.2 $\pm$ 24.2
BHA at			
50 mg/kg	7.2 $\pm$ 2.3	4.53 $\pm$ 0.37	16.4 $\pm$ 8.9
100 mg/kg	7.7 $\pm$ 1.4	4.66 $\pm$ 0.46	14.5 $\pm$ 9.6
250 mg/kg	6.4 $\pm$ 1.8 ^a	4.67 $\pm$ 0.27	18.3 $\pm$ 11.5
500 mg/kg	3.8 $\pm$ 4.8 ^b	4.73 $\pm$ 0.34 ^a	20.6 $\pm$ 12.6

Values are mean $\pm$ S.D. (n = 11).^a Significantly different from vehicle at p < 0.05.^b Significantly different from vehicle at p < 0.01.

Uterine weight and uterine epithelial cell height after the treatment of BHA in immature female rats

Groups	Uterine weight ^a		Uterine epithelial cell height (μ m)
	Absolute weight (mg)	Relative weight (wet, g/100 g)	
Vehicle (corn oil)	35.73 $\pm$ 4.65	0.0822 $\pm$ 0.0129	8.2 $\pm$ 0.6
17 β -estradiol (20 μ g/kg)	83.22 $\pm$ 10.03 ^c	0.2387 $\pm$ 0.0321 ^c	16.2 $\pm$ 2.1 ^c
BHA at			
50 mg/kg	29.55 $\pm$ 4.02 ^b	0.0679 $\pm$ 0.0097 ^c	8.0 $\pm$ 0.9
100 mg/kg	30.88 $\pm$ 3.36 ^c	0.0683 $\pm$ 0.0071 ^c	8.4 $\pm$ 0.1
250 mg/kg	28.17 $\pm$ 3.84 ^c	0.0666 $\pm$ 0.0072 ^c	8.3 $\pm$ 0.1
500 mg/kg	28.04 $\pm$ 4.30 ^c	0.0699 $\pm$ 0.0117 ^c	8.7 $\pm$ 0.8

Values are mean $\pm$ S.D. (n = 11).^a With fluid.^b Significantly different from vehicle at p < 0.05.^c Significantly different from vehicle at p < 0.01.Uterine and vagina weight after the treatment of 17 β -estradiol with or without BHA in immature female rats

Groups	Uterus		Vagina	
	Weight (mg) (blotted)	Relative weight (wet, g/100 g) (blotted)	Absolute weight (mg)	Relative weight (wet, g/100 g)
Vehicle (corn oil)	35.73 $\pm$ 4.65	0.080 $\pm$ 0.010	32.43 $\pm$ 2.03	0.073 $\pm$ 0.005
E2 (2 μ g/kg)	83.22 $\pm$ 10.03 (52.46 $\pm$ 4.53)	0.181 $\pm$ 0.016 (0.118 $\pm$ 0.009)	74.35 $\pm$ 7.94	0.167 $\pm$ 0.014
E2 + BHA (2 μ g/kg + 50 mg/kg)	87.38 $\pm$ 6.65 (56.29 $\pm$ 4.53)	0.186 $\pm$ 0.007 (0.126 $\pm$ 0.013)	78.89 $\pm$ 8.79	0.176 $\pm$ 0.018
E2 + BHA (2 μ g/kg + 500 mg/kg)	70.54 $\pm$ 13.01 ^a (43.93 $\pm$ 6.28 ^b)	0.169 $\pm$ 0.018 ^a (0.106 $\pm$ 0.011)	63.39 $\pm$ 8.57 ^b	0.152 $\pm$ 0.011 ^a

Values are mean $\pm$ S.D. (n = 7). E2; 17 β -estradiol.^a Significantly different from E2 at p < 0.05.^b Significantly different from E2 at p < 0.01.Ref.: Kang *et al.*, 2005; Anses, 2024**SCCS remark on the uterotrophic assay by Kang *et al.*, 2005**

After treatment with BHA, absolute and relative uterine weights were statistically significantly lower when compared to vehicle controls indicating an effect of BHA on estrogenic modality which is supported by an inhibition of 17 β -estradiol-stimulated increase of relative and

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absolute uterine and vaginal weight (uterus -15% relative and -7% absolute and vaginal -15% absolute and -9%, relative compared to E2 alone increase) at 500 mg/kg BHA + 2 µg/kg estradiol. However, it is of note that after treatment with BHA there was no dose-dependency in the reduction of absolute and relative uterine weights and the uterine epithelial cell height values were the same for the controls and the treated animals. Also, there are some deficiencies in reporting and methodology, e.g. with respect to statistics performed and phytoestrogen content in food (acceptable level of genistein equivalents exceeded).

Hershberger Assay

In the same study from the open literature, (anti-) androgenic properties of BHA (2(3)-tert-butyl-4-hydroxyanisole [25013-16-5], 99.9%) were investigated the Hershberger assay. BHA was dissolved in corn oil to prepare intended dosages of 50, 100, 250 and 500 mg/kg bw. 42 d old rats were castrated by removing testis and epididymis and recovered for 8 days. BHA and vehicle were administered via gavages to 51 d old castrated rats at 4 ml/kg bw daily for 10 days. Testosterone propionate (TP) (0.4 mg/kg bw) was dissolved in corn oil and administered by s.c. injection into the dorsal surface, caudal to the nape of neck at the volume of 0.4 ml/kg. Body weight and consumption of feed and water were recorded daily. Approximately 24 h after final treatment, androgen-dependent accessory sex glands or organs, ventral prostate (VP), seminal vesicle with coagulating glands (CV/CG), glans penis (GP), levator ani plus bulbocavernosus muscle (LABC), Cowper's gland (CG) were carefully removed and weighed. Especially for ventral prostate gland, the organ weight was measured again after fixation in 10% neutral formal-buffered for overnight. Liver, kidney, adrenal gland and thyroid gland were also weighed. Body weight gain, feed and water consumption and contents of testosterone and thyroxine in serum were not significantly changed by the treatment of BHA and there were no indications for toxicity. Body weight gain was increased by TP and TP with 250 mg/kg bw BHA in castrated rats in the absence of altered feed and water consumption. Treatment with BHA and TP caused statistically significant increase of ventral prostate weight in the 250 mg/kg BHA group but after fixation TP-stimulated ventral prostate weight was unchanged. In this Hershberger assay, indications for (anti-) androgenic activity of BHA were not observed.

SCCS comment on level 3 assays

The available information indicates some estrogenic activity of BHA.

Level 4: *in vivo* adverse effects data

See study Jeoung *et al.*, 2005, described in section reproductive toxicity.

SCCS overall conclusion on endocrine activity

Available information indicated weak anti-estrogenic activity of BHA. Based on *in vivo* studies of limited quality, some effects on sperm pointing to antiandrogenic modalities can be seen.

The SCCS is of the view that the overall evidence is not robust enough to clearly suggest an endocrine mode of action of BHA. However, there are indications that it may have antiestrogenic activity at high doses. Furthermore, some effects on sperm were described in an *in vivo* one generation reproductive toxicity study.

However, neither of these studies is considered sufficiently reliable to identify a point of departure from endocrine mode of action for risk assessment.

3.5 SAFETY EVALUATION (INCLUDING CALCULATION OF THE MoS)

As indicated in section 3.3.1 Function and Uses, the intended use concentrations are anticipated to be low based on a report from CIR (2023) and based on information obtained from IKW.

As communicated to the SCCS, use concentrations might not be higher than 0.07% in final Cosmetic Products. Therefore, this concentration of 0.07% will be used for safety calculation.

As explained in section 3.4.4 (Repeated dose toxicity), a NOAEL of 50 mg/kg bw/d is used as a PoD for risk assessment. As explained in section 3.2.1 (Dermal/percutaneous absorption), a default of 50 % is used as dermal absorption percentage.

No correction is made for oral absorption (see section 3.2.)

As explained in section 3.3.2, SEDs of **0.082** mg/kg bw/d (for leave-on and rinse-off products combined), **0.0029** mg/kg bw/d for rinse-off products and **0.079** mg/kg bw/d for leave-on products were calculated.

Safety calculations:

Leave-on and rinse-off products combined:

SED: 0.082 mg/kg bw/d

NOAEL (not adjusted): 50 mg/kg bw/d

Margin of Safety (NOAEL/SED)	adjusted NOAEL/SED	=	610
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Leave-on products alone:

SED: 0.079 mg/kg bw/d

NOAEL (not adjusted): 50 mg/kg bw/d

Margin of Safety (NOAEL/SED)	adjusted NOAEL/SED	=	632
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Rinse-off products alone:

SED: 0.0029 mg/kg bw/d

NOAEL (not adjusted): 50 mg/kg bw/d

Margin of Safety (NOAEL/SED)	adjusted NOAEL/SED	=	17251
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3.6 DISCUSSION

Physicochemical properties

BHA is composed of two isomers and appears as a white solid with a reported water solubility of 610 mg/l, an experimentally determined partition coefficient log Pow of 2.82 at 25°C and a molecular weight of 180.24 g/mol. It is stable under normal ambient and anticipated storage and handling conditions of temperature and pressure.

Toxicokinetics

The oral absorption of BHA is at least 86% in rats, 44% in dogs, and 32–100% in humans. The majority of the dose is excreted in the urine of both humans and rats during the first two days after oral administration (rat: approx. 49%, human: approx. 39%), primarily in conjugated form and as the demethylation product TBHQ or its conjugates (rat 8%, human 9%). Approximately 32% of free BHA is found in rat faeces; it is not detectable in humans. The plasma half-life of free BHA is less than one hour in both humans and rats. Excretion is largely complete after four days. BHA passes to rat fetuses to a small extent. Due to the high absorption rates observed in humans and animals, no correction for oral absorption will be made.

Exposure

The intended use concentration in cosmetics is up 0.07%.

Toxicological Evaluation

Irritation and corrosivity

BHA might be considered as slightly irritating to the skin. Reliable information on eye irritation potential of BHA is not available. However, concentration of BHA in cosmetic products is expected to be low and thus unlikely to result in eye irritation.

Skin sensitisation

There is no experimental data from animal studies or New Approach Methods (NAMs) on skin sensitisation potential and potency of BHA. The compilations of CIR and MAK provide clinical evidence that BHA can cause allergic contact dermatitis in humans. Based on these reports, the SCCS regards the risk of skin sensitisation from exposure to BHA in cosmetics negligible.

Acute toxicity

The available information indicates that BHA is of low acute oral and dermal toxicity.

Repeated dose toxicity

After oral administration, forestomach and liver were identified as target organs in oral repeat-dose studies in various animal species of different durations (see Annex I). The totality of these studies point to an NOAEL of 50 mg/kg bw/d based on liver effects. Systemic effects were not observed after repeated dermal administration of BHA.

Reproductive and developmental toxicity

The reproductive and developmental toxicity of BHA was investigated in a variety of *in vivo* studies in different species. Often, studies were quite old and not performed according to modern standards. Studies indicate effects on fertility (sperm), developmental neurobehavioural effects, indications for malformations. However, neither study was considered sufficiently robust to justify the selection of a PoD for risk assessment.

Mutagenicity / genotoxicity

Based on the observations from *in vitro* and *in vivo* studies performed with BHA, there is a concern with respect to a clastogenic potential. However, the positive findings in *in vitro* chromosomal aberration tests performed as well as the positive finding from an *in vivo* comet assay have to be put into perspective. Positive *in vitro* findings were only observed in the

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presence of S9 indicating that metabolism might be required to exert clastogenicity. It is known from literature, that BHA and its metabolites (e.g. tert-butylhydroquinone) can lead to formation of reactive oxygen species in the presence of liver microsomes or that BHA could be oxidized to produce free radicals. As described for other antioxidants, in living organisms, the occurrence of clastogenicity will be dependent from (i) metabolism, (II) exhaustion of antioxidant capacity and last but not least from the amounts (systemically) available at the target site.

With respect to the positive findings observed in an *in vivo* comet assay performed in mice, it is of note that positive findings were observed at high doses (500 and 1000 mg/kg). Such doses might indeed have exhausted antioxidant capacities in the target cells, but more importantly, when looking at the toxicological effects of BHA from a more holistic point of view, it should be kept in mind that liver toxicity (after repeated administration) already starts at much lower doses.

Carcinogenicity

The SCCS recognises that BHA induced forestomach tumours in 3 different rodent species. This tumour type is considered not relevant for humans. From a mechanistic point of view, the available information indicated that tumour formation might be due to formation of radicals and/or reactive oxygen species at high doses causing local cytotoxicity and finally leading to tumours. Based on the anticipated low intended use concentration in cosmetic, carcinogenicity is not considered as concern with respect to cosmetic uses.

Special investigations – Endocrine Disruption

Available information indicated weak anti-estrogenic activity of BHA. Based on *in vivo* studies of limited quality, some effects on sperms pointing to antiandrogenic modalities can be seen. The SCCS is of the view that the overall evidence is not robust enough to clearly suggest an endocrine mode of action of BHA. However, there are indications that it may have antiestrogenic activity at high doses. Furthermore, some effects on sperms were described in an *in vivo* one generation reproductive toxicity study. However, neither of these studies is considered sufficiently reliable to identify a point of departure from endocrine mode of action for risk assessment.

Scientific advice on the safety of Butylated Hydroxyanisole (BHA) as a substance with potential endocrine
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4. CONCLUSION

In light of technical and scientific progress, available scientific literature, relevant in silico tools and the SCCS' expert judgement and taking under consideration in particular the concerns related to potential endocrine disrupting properties, the SCCS is requested:

1. *to identify and justify whether there are specific concerns regarding the use of BHA in cosmetic products;*

The SCCS considers the use of BHA in leave-on and rinse off cosmetic products up to a concentration of 0.07% as safe.

2. *to highlight if there is a potential risk for human health from the use of BHA in cosmetic products;*

This SCCS Opinion considered only dermal use and hence, is applicable for the use of BHA in dermally applied products, and not in oral care products or cosmetic products which may lead to exposure of the end-user's lungs by inhalation.

The SCCS mandate does not address environmental aspects. Therefore, this assessment has not covered the safety of BHA for the environment.

5. MINORITY OPINION

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7. GLOSSARY OF TERMS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – from page 158

8. LIST OF ABBREVIATIONS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – from page 158

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1

2 **Annex I**

3 Repeated dose toxicity studies performed with BHA

4

Species, Strain, number of animals per group	Exposure	Findings	Reference
Rat, Wistar, 6 ♂	3 and 7 days 0; 1,5% BHA (1500 mg/kg bw) and 0,2% Acetylsalicylic acid, diet	1500 mg/kg bw: rel. liver weight ↑, forestomach: hyperplasia	Schilderman <i>et al.</i> 1995
	0; 1,5%BHA (1500 mg/kg bw), diet	1500 mg/kg bw: rel. liver weight ↑, forestomach: hyperplasia	
	0; 0,2% Acetylsalicylic acid	bw ↓ (3 days), abs. liver weight ↓ forestomach: slight hyperplasia (1/6; controls: 1/6)	
Rat, Wistar, 6 ♂	14 days 0; 1,5% BHA (1500 mg/kg bw/d) and 0,2% Acetylsalicylic acid, diet	1500 mg/kg bw: liver: abs. liver weight ↑, 8-oxodeoxyguanosin ↑ forestomach: hyperplasia, inflammation,	Schilderman <i>et al.</i> 1995
	0; 1,5%BHA (1500 mg/kg bw/d), diet	1500 mg/kg bw: forestomach: hyperplasia, inflammation	
	0; 0,2% acetylsalicylic acid	forestomach: hyperplasia	
Rat, Sprague Dawley, 4 ♂	1, 2, 4 and 6 days, 0, 500 mg BHA /kg bw/d, gavage	500 mg/kg bw: Na ⁺ -excretion via urine decreased (only renal function examined)	Ford <i>et al.</i> 1980 a, 1980 b
Rat, Carworth, je 4 ♂, ♀	7 days, 0, 50, 100, 200, 500 mg BHA/kg bw/d, gavage	50 mg/kg bw/d: NOAEL 100 mg/kg bw/d: ♂: rel. liver weight ↑ 200 mg/kg bw/d: ♀: rel. liver weight ↑	Feuer <i>et al.</i> 1965
Rat, Carworth, je 4–6 ♂, ♀	7 day, 0; 500 mg BHA/kg bw/d gavage, 14 and 28 day recoveries	500 mg/kg bw/d: ♂: glucose-6- phosphatedehydrogenase after 14 days ↑, reversible.	Feuer <i>et al.</i> 1965
Rat, F344, 5 ♂	9 days, 0; 0,1; 0,25; 0,5; 1; 2% BHA, (ca. 100, 250, 500, 1000, 2000 mg/kg bw/d), diet	from 500 mg/kg bw/d: proliferation epithelium forestomach	Nera <i>et al.</i> 1984

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Species, Strain, number of animals per group	Exposure	Findings	Reference
Rat, Wistar, 5 ♂, ♀ per dose	1, 2 and 4 wks, 0, 2% BHA (ca. 2000 mg/kg bw/d), diet After treatment: 4 wk normal diet	At 2000 mg/kg bw/d: Forestomach: slight hyperplasia And hyperkeratosis after 2 wks; Slight hyperplasia after 4 wks	Altmann <i>et al.</i> , 1986

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Rat, F344, 5 ♂	4 wks, 0, 2% BHA (ca. 2000 mg/kg bw/d), diet	At 2000 mg/kg bw/d: decreased bw gain increased liver weight forestomach hyperplasia	Hirose <i>et al.</i> 1987b 2 3 4
Ratte, Norway Hooded, 6 ♂, ♀ per dose	6 wks, 0; 0,1; 0,2; 0,3; 0,4; 0,5% BHA (ca. 100, 200, 300, 400, 500 mg/kg bw/d), diet	From 100 mg/kg bw/d: increased Serumcholesterine rel. weights of liver and adrenal glands	Johnson and Hewgill 1961
Rat, F344, 25 ♂	13 wks, 0; 0,1; 0,25; 0,5; 2% BHA (ca. 75, 188, 375, 1500 mg/kg bw/d), diet	At 1500 mg/kg bw/d: Forestomach: damaged epithelium, hyperkeratoris, Parakeratosis Parakeratose	Iverson <i>et al.</i> , 1985a
Rat, Wistar, 10–15 ♂, ♀ per dose	13 wks, 0; 0,125; 0,5; 2% BHA (ca. 94, 375, 1500 mg/kg bw/d), diet	from 94 mg/kg bw/d: forestomach hyperplasia	Altmann <i>et al.</i> 1986
Rat, Wistar, control: 5 ♂, ♀; dosed animals: 15 ♂, ♀	13 wks, 0, 2% BHA (ca. 1500 mg/kg bw/d), diet after treatment: 4 wk normal diet	1500 mg/kg bw/d: forestomach hyperplasia	Altmann <i>et al.</i> 1986
Rat, Wistar, 10 ♂, ♀ per dose	13 wks, 0; 0,025; 0,125; 2%; 5% BHA (ca. 19, 94, 1500, 3750 mg/kg KG bw/d), initially in peanut oil, later as food admixture	3750 mg/kg bw/d: pronounced forestomach hyperplasia; in 3 animals: papillar hyperplasia	Altmann <i>et al.</i> 1986

Species, Strain, number of animals per group	Exposure	Findings	Reference
Rat, Wistar, 5 ♂, ♀ per dose	13 wks, 0; 0,025; 0,125; 2% BHA (ca. 19, 94, 1500 mg/kg bw/d), initially in peanut oil, later as food admixture, after treatment: 4 wk normal diet	1500 mg/kg bw/d: ♀: slight forestomach hyperplasia (1/5)	Altmann <i>et al.</i> 1986
Rat, Carworth, each 24 ♂, ♀	16 wks, 0; 0,1% BHA (ca. 75 mg/kg bw/d), diet	75 mg/kg bw/d: ♀: increased rel. weights of liver and adrenal glands Increased urinary excretion of ascorbic acid↑, ♂: decreased bw	Gaunt <i>et al.</i> 1965
Rat, F344, 10 ♂	24 wks, 2% BHA (ca. 1500 mg/kg bw/d), diet	dosed animals: forestomach: thickened epithelium, hyperplasia, papilloma, proliferation of epithelium and basal cells	Masui <i>et al.</i> 1986 a
Rat, F344, 20 ♂	24 wks, 0, 2% BHA (ca. 1500 mg/kg bw/d), diet, after treatment: 72 wks normal diet	dosed animals: forestomach: thickened epithelium, hyperplasia, papilloma, proliferation of epithelium and basal cells	Masui <i>et al.</i> 1986 a

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Rat, Wistar, 20 (sex not specified)	6 mo, 0, 2% BHA (ca. 1500 mg/kg bw/d), diet, after treatment: 2 mo normal diet	dosed animals: forestomach: hyperplasia, hyperkeratosis	Altmann <i>et al.</i> 1986
Rat, Wistar, 30 (sex not specified)	12 mo, 0; 2% BHA (ca. 1000 mg/kg bw/d), diet, after treatment: 2 mo normal diet	dosed animals: forestomach: hyperplasia, hyperkeratosis, parakeratosis, dysplasia, papilloma, invasive	Altmann <i>et al.</i> 1986
Rat, Wistar, 28 (sex not specified)	15 mo, 0, 2% BHA (ca. 1000 mg/kg bw/d), diet, after treatment: 7 mo normal diet	dosed animals: forestomach: pronounced hyperplasia: 1/14; hyperkeratosis: 1/14 close to esophagus: moderate hyperplasia: 1/14; hyperkeratosis 5/14	Altmann <i>et al.</i> 1986

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Species, Strain, number of animals per group	Exposure	Findings	Reference
Rat, Norway Hooded; Each 20 ♂, ♀	8 mo 0; 0,001; 0,1; 0,5% BHA 250 mg/kg (ca. 0.5, 50, 250 mg/kg bw/d), diet	NOAEL: 50 mg/kg bw/d 250 mg/kg bw/d: increased liver weight (investigated tissues: liver, kidney, spleen, testes)	Brown et al., 1959
Rat, F344 50 ♂	104 wk 0; 0,125; 0,25; 0,5; 1; 2% BHA (55, 110, 230, 428, 1323 mg/kg bw/d), diet	NOAEL: 55 mg/kg bw/d from 110 mg/kg bw/d: decreased bw gain; forestomach hyperplasia at 230 and 428 mg/kg bw/d: hyperplasia of liver (not dose- dependent)	Ito et al., 1986a
Rat, F344 up to 31 ♂	104 wk 0; 0,4% BHA (0 and ca. 200 mg/kg bw/d), diet	At 200 mg/kg bw/d: Rel. liver and kidney weight increased	Hirose et al., 1997
Rat, F344 27 ♂	110 wk 0 and 1200mg BHA/kg diet (ca. 600 mg/kg bw/d)	At 600 mg/kg bw/d: liver weight affected, liver effects, forestomach hyperplasia	Williams et al., 1990b
Rat, F344 50 ♂	2 years 0, 1% and 2% BHA (ca 500 and 1000 mg/kg bw/d), diet	From 500 mg/kg bw/d: forestomach hyperplasia (only forestomach and stomach investigated)	Masui et al., 1986b
Mouse, CD-1 30 ♀	10 d 0; 0,75% BHA (0 and 1500 mg/kg bw/d), diet	At 1500 mg/kg bw/d: Enlarged liver,; rel. liver weight increased, increased protein, induction of microsomal enzymes	Cha and Bueding, 1979
Mouse, NMRI 10 ♂	28 d 0, 1000 mg/kg bw/d, gavage	At 1000 mg/kg bw/d: Macroscopical damage in forestomach (only stomach and forestomach investigated)	Altmann <i>et al.</i> , 1986
Syrian Golden Hamster 10 ♂	28 d 0; 2% BHA (ca. 1000 mg/kg bw/d), diet	At 1000 mg/kg bw/d: Mild hyperplasia and hyperkeratosis (only forestomach and stomach investigated)	Altmann <i>et al.</i> , 1986

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Syrian Golden Hamster 15 ♂	20 wk 0; 0,5% BHA (ca. 250 mg/kg bw/d), diet	At 250 mg/kg bw/d: Forestomach hyperplasia; no histopathological changes in liver, kidney, urinary bladder	Hirose <i>et al.</i> , 1986
Syrian Golden Hamster 15 ♂	20 wk 0; 0,5% Metabolite TBHQ (ca. 250 mg/kg bw/d), diet	At 250 mg/kg bw/d: forestomach hyperplasia;	Hirose <i>et al.</i> , 1986
Syrian Golden Hamster 15 ♂	24 wk 0; 1% BHA (ca 500 mg/kg bw/d), diet	At 500 mg/kg bw/d: inflammation, hyperkeratosis, hyperplasia and papilloma in forestomach	Lam and Garg, 1991
Guinea pig, 10 ♂	28 d, 0, 1000 mg BHA/kg bw/d, gavage	At 1000 mg/kg bw/d: No effects in stomach (only stomach investigated)	Altmann <i>et al.</i> 1986
Rabbit, White New Zealand, 13 (sex not specified)	6 d, 1000 mg BHA/animal (ca. 333 mg/kg bw/d), gavage	At 333 mg/kg bw/d: 10-fold increase of sodium excretion, 20-fold increase potassium excretion; increased excretion of aldosterone	Denz and Llauro, 1957
Dog, Beagle, 3 ♂, 3–4 ♀	6 mo, 0; 0,25; 0,5; 1% BHA (♂ + rel. liver weight increased by 0, 54, 111, 219 mg/kg bw/d; ♀ 62, 112, 231 mg/kg bw/d), diet	from 54/62 mg/kg bw/d: ♂: abs. 27–50%; reduced ratio albumin/globulin; ♀: rel. liver weight increased; reflux of bile into stomach lumen; ♂/♀: decreased body weight gain; from 111/112 mg/kg bw/d: ♀: increased leucine-aminopeptidase; at 219/231 mg/kg bw/d ♂, ♀: increased alkaline phosphatase	Tobe <i>et al.</i> 1986
Dog, Beagle 29 ♂, 30 ♀	180 d, 0; 1; 1,3% BHA (ca. 200, 250 mg/kg bw/d), diet	from 200 mg/kg bw/d: food consumption, body weight decreased, liver weight increased, no particular findings in esophagus, stomach	Ikeda <i>et al.</i> , 1986
Dog, Cocker-Spaniel 1 ♂, 3 ♀	15 mo 0, 5, 50, 250 mg/kg bw/d, diet	NOAEL: 50 mg/kg bw/d at 250 mg/kg bw/d: food consumption + body weight gain decreased: urinary glucuronic acid decreased; degeneration of liver cells, infiltration of granulocytes, haemosiderin in Kupffer cells; 13 further tissues (including stomach) without effects	Wilder <i>et al.</i> , 1960
Monkey, Cynomolgus, 8 (sex not specified)	85 d, 0, 125, 500 mg/kg bw/d, gavage; after 20 d: highest dose reduced to 250 mg/kg bw due to emesis	from 125 mg/kg bw/d: abs. + rel. liver weight dose-dependently increased; Ethoxyresorufin-deethylase activity decreased	Iverson <i>et al.</i> 1985 b, 1986
Monkey, Rhesus, 3 young (1 mo), 2–3 juvenile	4 wk, 7 d/wk 0, 50, 500 mg/kg bw/d, gavage	At 50 mg/kg bw/d: rel. liver weight increased (statistically not significant); at 500 mg/kg bw/d: rel. liver weight statistically significantly increased. Liver: enlargement, proliferation of endoplasmic reticulum, fat droplets in cytoplasm; fragmentation of nucleus, decreased cholesterol Cholesteringehalt ↓	Allen und Engblom 1972; Brannen <i>et al.</i> 1973

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